CAREERLY-001 — Authentication Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Authentication (Register, Login, Password Reset)
Flow ID:       CAREERLY-001
Trigger:       User visits the platform for the first time, or returns to log in, or requests a password reset
Entry Point:   Landing page / Auth screen
Exit Point:
  Register     → Home page (profile + resume set up)
  Login        → Home page
  Password     → Profile page
Related Flows: CAREERLY-005 (Profile), CAREERLY-002 (Home Page)
```

1.2 Description

This flow covers all three authentication routes for Careerly: registration, login, and password reset. It exists to securely onboard new job seekers and return authenticated users to the platform. The business value is establishing a verified user identity tied to a resume and preferences, which every other feature in the platform depends on. Registration is the critical path — it is not complete until the user uploads a resume and sets their preferences, as these are required for job recommendations to function. Login and password reset are standard credential flows with OTP-based email verification.

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Job Seeker	Human	Provides credentials, verifies OTP, uploads resume, sets preferences
System	Automated	Validates inputs, generates and sends OTP, creates user record, issues auth token
Email Service	Third-Party	Delivers OTP emails

1.4 Step-by-Step Bullet Points

Route 1 — Register

- Job Seeker — navigates to the Register screen and enters email, password, and confirm password
- System — validates all fields (format, strength, match)
  ↳ if validation fails: highlights invalid fields with error messages, does not proceed
- System — checks if email is already registered
  ↳ if duplicate: shows "An account with this email already exists" message, does not proceed
- System — creates a user record with status = `UNVERIFIED`
- System — generates a 6-digit OTP, stores it with a 10-minute expiry, and dispatches a Django Task to send the OTP email
- Job Seeker — is redirected to the OTP verification screen
- Job Seeker — enters the 6-digit OTP
- System — validates the OTP (correct, not expired)
  ↳ if OTP is wrong: shows "Invalid code. Please try again." — allows retry
  ↳ if OTP is expired: shows "Code expired. Request a new one." — shows resend button
  ↳ if OTP attempts exceed 5: locks OTP entry for 15 minutes
- System — marks user status = `VERIFIED` , generates auth token
- Job Seeker — is redirected to the Preferences + Resume setup screen
- Job Seeker — selects job preferences (job titles of interest, locations, job types)

- Job Seeker — uploads their resume (PDF, required — cannot skip)
  ↳ if user tries to skip resume: shows "Resume is required to continue" — blocks navigation
- System — validates resume file (type, size)
  ↳ if invalid file type: shows "Only PDF files are accepted"
  ↳ if file too large: shows "File size must be under 5MB"
- System — saves preferences and resume, links them to the user profile
- System — dispatches Celery Task to parse resume in background (AI extraction of skills, titles, etc.)
- Job Seeker — is redirected to the Home page
- System — issues JWT access token + refresh token, stored client-side

Route 2 — Login

- Job Seeker — navigates to the Login screen and enters email and password
- System — validates fields (both required, non-empty)
  ↳ if fields are empty: shows "Please fill in all fields"
- System — looks up email, verifies password hash
  ↳ if email not found or password wrong: shows "Incorrect email or password" (intentionally vague — no account enumeration)
  ↳ if user status = `UNVERIFIED` : shows "Please verify your email first" — offers resend OTP
  ↳ if account is suspended: shows "Your account has been suspended. Contact support."
- System — issues JWT access token + refresh token
- Job Seeker — is redirected to the Home page

Route 3 — Password Reset

- Job Seeker — navigates to the Forgot Password screen and enters their email
- System — validates email format
  ↳ if invalid format: shows "Please enter a valid email address"
- System — checks if email exists
  ↳ if email not found: shows the same success message anyway (prevents account enumeration)
- System — generates a 6-digit OTP, stores it with a 10-minute expiry, dispatches Django Task to send email
- Job Seeker — is redirected to OTP verification screen
- Job Seeker — enters the OTP
- System — validates OTP (correct, not expired)
  ↳ if wrong: shows "Invalid code. Please try again."
  ↳ if expired: shows "Code expired." — shows resend button (disabled for 60 seconds after each send)
  ↳ if attempts exceed 5: locks for 15 minutes
- System — issues a short-lived password reset token (valid 15 minutes, single use)
- Job Seeker — is redirected to the New Password screen (2 inputs: new password, confirm password)
- Job Seeker — enters and confirms new password
- System — validates password strength and match
  ↳ if mismatch: shows "Passwords do not match"
  ↳ if weak: shows strength requirements
- System — hashes and saves the new password, invalidates the reset token, invalidates all existing sessions
- Job Seeker — is redirected to the Profile page (logged in automatically)

1.5 Validations

Input Validations

Field	Rule	Error Message
Email	Required, valid email format	"Please enter a valid email address"
Password (Register)	Required, min 8 chars, at least 1 uppercase, 1 number, 1 special character	"Password must be at least 8 characters and include an uppercase letter, a number, and a special character"
Confirm Password	Must match password field exactly	"Passwords do not match"

Field	Rule	Error Message
OTP	Required, exactly 6 digits, numeric only	"Please enter the 6-digit code"
Resume	Required, PDF only, max 5MB	"Only PDF files are accepted" / "File size must be under 5MB"
New Password	Same rules as registration password	Same message
Job Preferences	At least 1 job title required	"Please select at least one job preference to continue"

Business Rule Validations

Rule	Condition	Behavior
No duplicate accounts	Email already registered	Block + "An account with this email already exists"
Resume is mandatory	User tries to proceed without uploading	Block navigation + show message
OTP expiry	OTP older than 10 minutes	Invalidate + prompt resend
OTP attempt limit	More than 5 wrong attempts	Lock entry for 15 minutes
Resend cooldown	OTP resend requested	Disable resend button for 60 seconds
Reset token single-use	Token already used	Reject + redirect to forgot password
Session invalidation on reset	Password changed	All existing JWT tokens are blacklisted
Unverified login	User logs in before verifying email	Block + offer resend OTP

Security Validations

Check	Details
Authentication	Not required for register/login/reset — these are public endpoints
Account enumeration prevention	"Email not found" and "Wrong password" return the same error message
Password hashing	bcrypt via Django's default hasher — never store plain text
OTP storage	Stored hashed, not plain text, with expiry timestamp
JWT	Access token: 15 min expiry. Refresh token: 30 days. Stored in HttpOnly cookies or secure storage
Reset token	Single-use, 15-minute expiry, invalidated on use or on new request
HTTPS	All auth endpoints must be HTTPS only

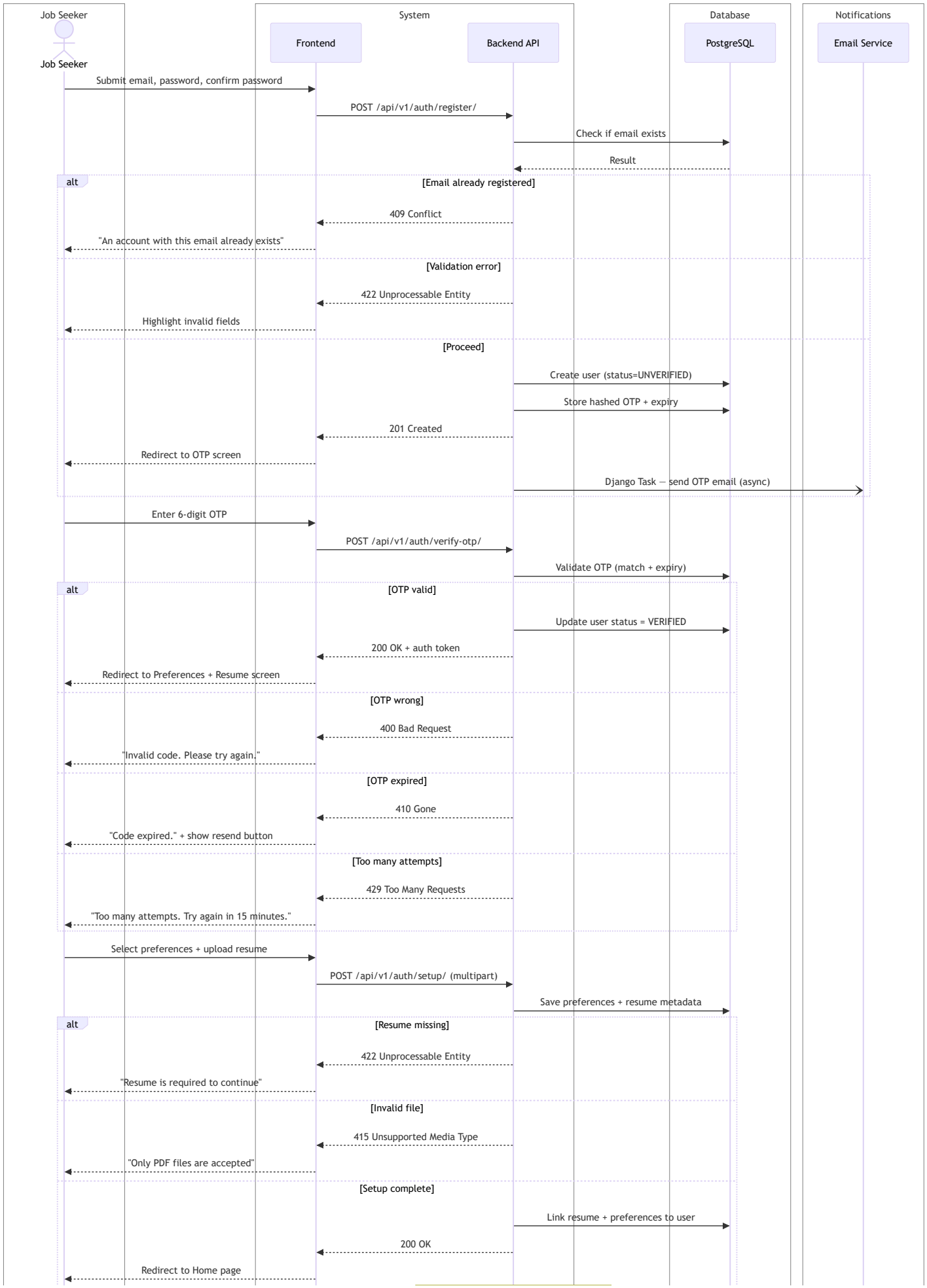
Error Handling

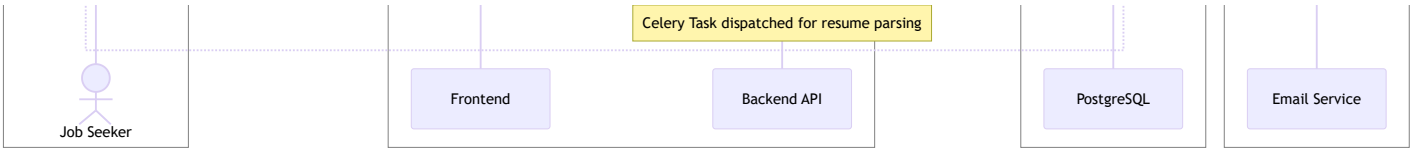
Scenario	System Response
Email service fails (OTP not sent)	Show "We couldn't send the code. Please try again." — do not create user record yet
Server error on register submit	Show generic error, preserve form data, allow retry
Resume upload network failure	Show "Upload failed. Please try again." — preserve other form data
Token refresh fails	Force logout, redirect to login with "Session expired" message
Timeout on OTP verification screen	Keep screen alive — OTP expiry is time-based, not session-based

PART 2 — TECHNICAL

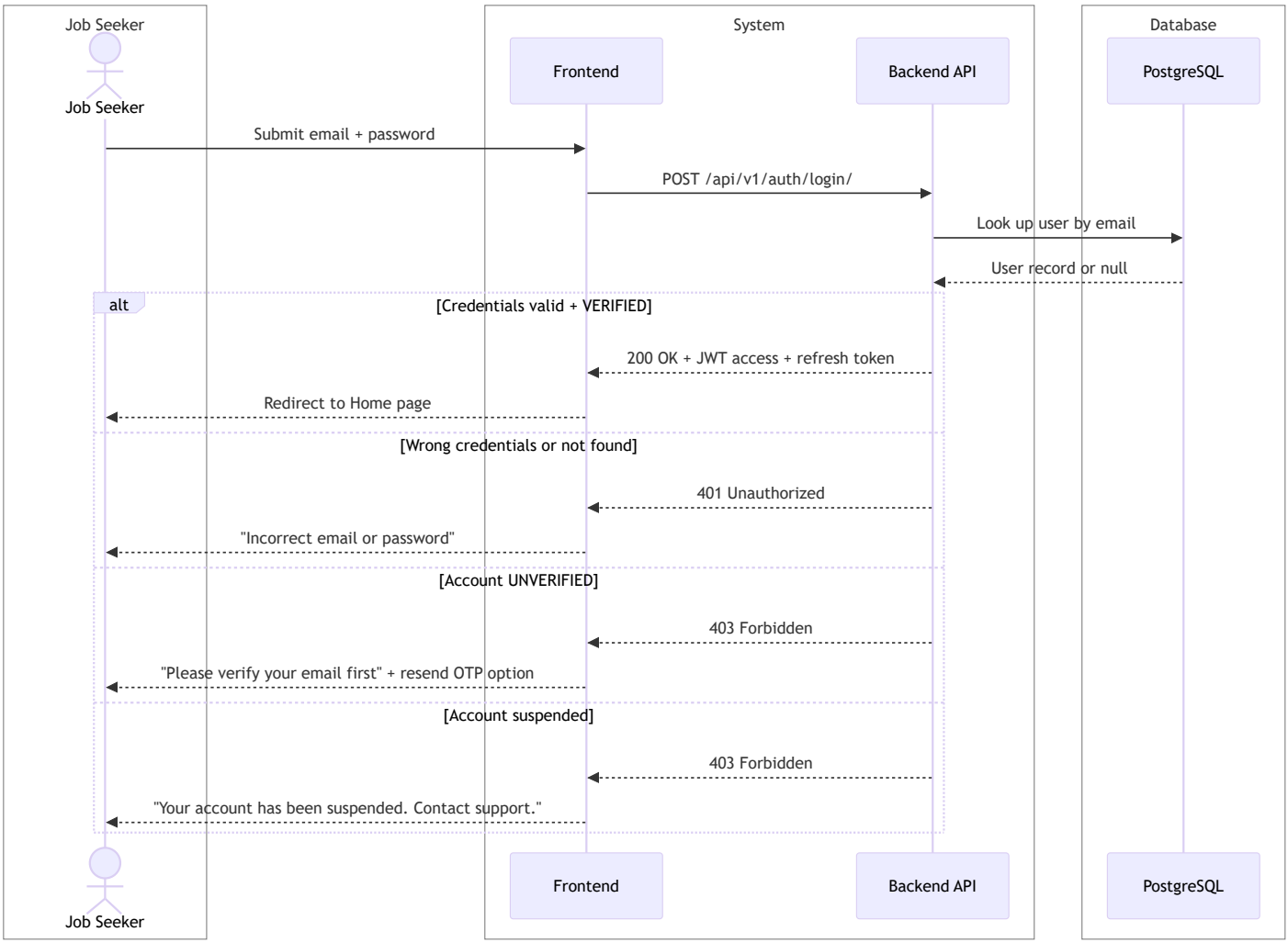
2.1 Diagrams

Sequence Diagram — Register

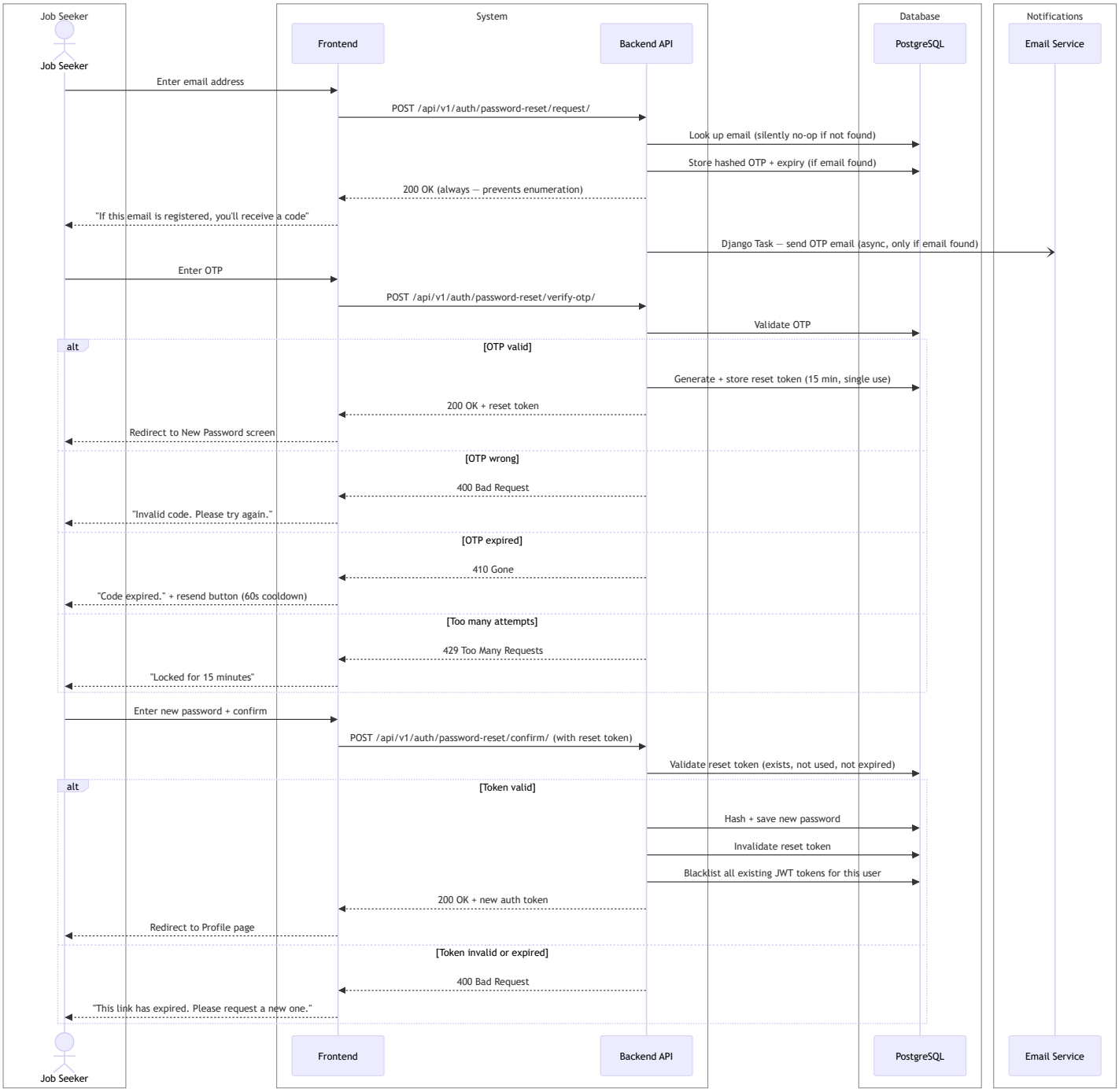

Job Seeker
System
Database
Notifications
Frontend
Backend API
PostgreSQL
Email Service
Job Seeker
Submit email, password, confirm password
POST /api/v1/auth/register/
Check if email exists
Result
alt
[Email already registered]
409 Conflict
"An account with this email already exists"
[Validation error]
422 Unprocessable Entity
Highlight invalid fields
[Proceed]
Create user (status=UNVERIFIED)
Store hashed OTP + expiry
201 Created
Redirect to OTP screen
Django Task — send OTP email (async)
Enter 6-digit OTP
POST /api/v1/auth/verify-otp/
Validate OTP (match + expiry)
alt
[OTP valid]
Update user status = VERIFIED
200 OK + auth token
Redirect to Preferences + Resume screen
[OTP wrong]
400 Bad Request
"Invalid code. Please try again."
[OTP expired]
410 Gone
"Code expired." + show resend button
[Too many attempts]
429 Too Many Requests
"Too many attempts. Try again in 15 minutes."
Select preferences + upload resume
POST /api/v1/auth/setup/ (multipart)
Save preferences + resume metadata
alt
[Resume missing]
422 Unprocessable Entity
"Resume is required to continue"
[Invalid file]
415 Unsupported Media Type
"Only PDF files are accepted"
[Setup complete]
Link resume + preferences to user
200 OK
Redirect to Home page



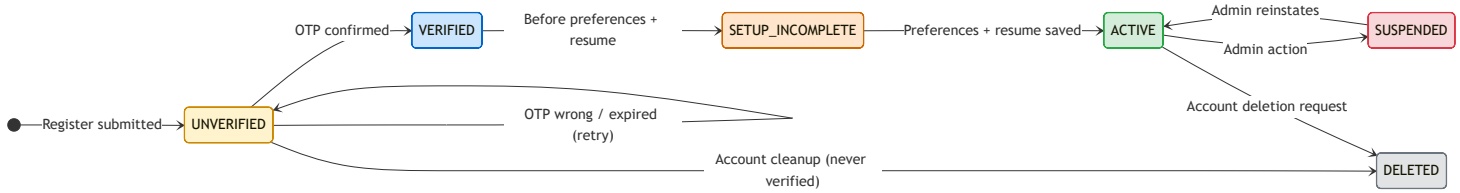
Sequence Diagram — Login



Sequence Diagram — Password Reset



State Diagram — User Account Status



2.2 Data Models

Model: `User`

Purpose: Core user account — authentication identity and status
Django app: `accounts`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK, auto-generated
`email`	`EmailField(unique=True)`	Yes	—	Unique, indexed — primary login identifier
`password`	`CharField(max_length=128)`	Yes	—	Django hashed password field — use `set_password()`
`status`	`CharField(choices=ACCOUNT_STATUS, max_length=20)`	Yes	`UNVERIFIED`	Enum: UNVERIFIED, VERIFIED, SETUP_INCOMPLETE, ACTIVE, SUSPENDED, DELETED
`is_admin`	`BooleanField`	No	`False`	True for admin users — controls admin dashboard access
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	Immutable creation timestamp
`updated_at`	`DateTimeField(auto_now=True)`	Auto	`now`	Auto-updated on save
`last_login`	`DateTimeField(null=True, blank=True)`	No	`None`	Updated on successful login

Model: `OTPRecord`

Purpose: Stores OTP codes for email verification and password reset
Django app: `accounts`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Owning user — delete OTPs when user deleted
`otp_hash`	`CharField(max_length=128)`	Yes	—	SHA-256 hash of the 6-digit OTP — never store plain
`purpose`	`CharField(choices=OTP_PURPOSE, max_length=20)`	Yes	—	Enum: EMAIL_VERIFICATION, PASSWORD_RESET
`expires_at`	`DateTimeField`	Yes	—	Set to `now + 10 minutes` on creation. Indexed.
`attempts`	`PositiveIntegerField`	No	`0`	Incremented on each wrong guess
`is_used`	`BooleanField`	No	`False`	Marked True after successful verification
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Model: `PasswordResetToken`

Purpose: Short-lived single-use token issued after OTP verification, used to authorize the actual password change
Django app: `accounts`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK — also used as the token value
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Cascade delete with user
`expires_at`	`DateTimeField`	Yes	—	`now + 15 minutes`
`is_used`	`BooleanField`	No	`False`	Invalidated on first use

Field	Django Field Type	Required	Default	Notes
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Model: `UserPreferences`

Purpose: Job seeker preferences set during onboarding, used for recommendations
Django app: `accounts`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`OneToOneField(User, on_delete=CASCADE)`	Yes	—	One preferences record per user
`job_titles`	`ArrayField(CharField(max_length=100))`	Yes	—	At least 1 required. Postgres ArrayField.
`locations`	`ArrayField(CharField(max_length=100))`	No	`[]`	Preferred job locations
`job_types`	`ArrayField(CharField(max_length=50))`	No	`[]`	Enum values: FULL_TIME, PART_TIME, REMOTE, CONTRACT, INTERNSHIP
`updated_at`	`DateTimeField(auto_now=True)`	Auto	`now`	—

2.3 Table Relationships & Logic

`User` is the root entity. Every other model in this flow hangs off it. When a `User` is deleted, all related `OTPRecord`, `PasswordResetToken`, and `UserPreferences` records are cascade-deleted.

`OTPRecord` can have multiple records per user over time (one per request). Before creating a new OTP, the system should mark all previous OTPs of the same `purpose` as `is_used=True` to prevent old codes from being valid. Query pattern: `OTPRecord.objects.filter(user=user, purpose=purpose, is_used=False).order_by('-created_at').first()`.

`PasswordResetToken` is a separate model from OTP intentionally — it has a different lifecycle. It is issued only after OTP verification succeeds, so there is no risk of someone using just an email to get a reset token. It is single-use and expires in 15 minutes.

`UserPreferences` is a `OneToOneField` — there is exactly one preferences record per user. It is created during the setup step, not at registration. A `post_save` signal on `UserPreferences` should update the user's `status` to `ACTIVE` once preferences and a resume both exist.

Status transition logic — the `User.status` field is the canonical source of truth for what the user is allowed to do:

- `UNVERIFIED` → cannot log in
- `VERIFIED` → can log in but will be redirected to setup if `UserPreferences` or resume is missing
- `SETUP_INCOMPLETE` → logged in but gated — only setup screen accessible
- `ACTIVE` → full access
- `SUSPENDED` → read-only or full block depending on admin decision

OTP rate limiting — `attempts` on `OTPRecord` is checked before validating. If `attempts >= 5`, return 429 without checking the code. On each wrong guess, increment `attempts`. On success, set `is_used=True`.

Resend cooldown — enforced by checking `created_at` of the most recent OTP. If `now - created_at < 60 seconds`, reject the resend request with 429.

2.4 API Endpoints

Method	Endpoint	Auth	Role	Request Body	Re
`POST`	`/api/v1/auth/register/`	No	—	`{email, password, confirm_password}`	`201`
`POST`	`/api/v1/auth/verify-otp/`	No	—	`{email, otp, purpose}`	`200 + tol`

Method	Endpoint	Auth	Role	Request Body	Re
POST	/api/v1/auth/resend-otp/	No	—	{email, purpose}	200
POST	/api/v1/auth/setup/	Yes	Job Seeker	multipart: {job_titles, locations, job_types, resume_file}	200
POST	/api/v1/auth/login/	No	—	{email, password}	200 + JW
POST	/api/v1/auth/logout/	Yes	Any	{refresh_token}	204
POST	/api/v1/auth/token/refresh/	No	—	{refresh_token}	200 + ne
POST	/api/v1/auth/password-reset/request/	No	—	{email}	200
POST	/api/v1/auth/password-reset/verify-otp/	No	—	{email, otp}	200 + re
POST	/api/v1/auth/password-reset/confirm/	No	—	{reset_token, new_password, confirm_password}	200 + JW

2.5 Developer Notes

🔵 Backend Developer (Django)

- Use `AbstractBaseUser + BaseUserManager` for the custom `User` model — do not use Django's default `User` since we need `email` as the login field instead of `username`.
- Use `djangorestframework-simplejwt` for JWT — configure `ACCESS_TOKEN_LIFETIME = timedelta(minutes=15)` and `REFRESH_TOKEN_LIFETIME = timedelta(days=30)`.
- Blacklist refresh tokens on logout and on password reset using simplejwt's token blacklist app — add `rest_framework_simplejwt.token_blacklist` to `INSTALLED_APPS`.
- OTP generation: use `secrets.randbelow(1000000)` zero-padded to 6 digits. Hash with `hashlib.sha256` before storing.
- Resume file upload: use `django-storages` with S3 (or local `FileField` for development). Store file path in the `Resume` model (defined in CAREERLY-005 Profile flow).
- `post_save` signal on `UserPreferences`: check if user also has at least one resume — if yes, set `user.status = ACTIVE`.
- Use `transaction.on_commit()` before dispatching Django Task for OTP email.
- Rate limiting on OTP endpoints: enforce at the model level (attempts counter) AND at the API level with `django-ratelimit`.
- For the setup endpoint, handle `multipart/form-data` — use DRF's `MultiPartParser`.
- Password reset token: use the record's UUID as the token. No need for a separate token string.

🟢 Frontend Developer (React)

- **Register screen**: email, password, confirm password fields. Show password strength indicator. Disable submit button until all fields pass client-side validation.
- **OTP screen**: 6 individual digit input boxes (auto-advance on each digit entry). Show countdown timer for expiry (10 min). Show resend button — disable it for 60 seconds after each send, display countdown.
- **Preferences + Resume screen**: multi-select for job titles and locations (searchable dropdown), checkbox group for job types. File upload zone for resume (drag-and-drop + click). Show file name after upload. Disable "Continue" button until resume is attached.

- **Login screen**: email + password. "Forgot password?" link routes to reset flow.
- **Password reset — New Password screen**: 2 fields (new password, confirm). Show same strength indicator as register.
- Store JWT access token in memory (not localStorage). Store refresh token in HttpOnly cookie or secure storage.
- On 401 responses: attempt silent refresh via `/api/v1/auth/token/refresh/` . If refresh fails, force logout.
- Redirect logic after login: check user `status` from the profile API — if `SETUP_INCOMPLETE` , redirect to setup screen instead of home.

🟡 Mobile Developer (Flutter)

- **OTP input**: use a `Row` of 6 `TextField` widgets with `maxLength=1` , `keyboardType=TextInputType.number` . Auto-focus next field on entry, auto-submit when 6th digit entered.
- **Resume upload**: use `file_picker` package — filter to `.pdf` and `.docx` only. Show file name and size after selection.
- **Token storage**: use `flutter_secure_storage` for both access and refresh tokens — never use `SharedPreferences` for tokens.
- **Silent refresh**: implement an HTTP interceptor (using `dio` interceptors) that catches 401 responses, attempts token refresh, then retries the original request.
- **Deep links**: register `careerly://auth/verify-otp` deep link so that if the user opens the OTP email on mobile, they are taken directly to the OTP screen.
- **Keyboard types**: email fields use `TextInputType.emailAddress` , password fields use `obscureText: true` with a visibility toggle.
- **Navigation**: use `pushReplacement` after login and after setup completion — user should not be able to navigate back to auth screens using the back button.

🟣 AI Engineer

- On successful setup (resume uploaded + preferences saved), a Celery task is dispatched to parse the resume.
- Input: resume file (PDF), user ID.
- Output: extracted fields — job titles, skills, years of experience, education level. Store as structured JSON on the `Resume` model.
- This parsed data is used in CAREERLY-002 (Home Page) for job recommendations.
- Latency: async — user does not wait for this. Show "We're setting up your recommendations" on first home page visit if parsing is still in progress.
- Fallback: if parsing fails or times out, keep the resume on file and retry once. If retry fails, flag the resume as `parse_status=FAILED` — recommendations will fall back to preferences-only matching until re-parsed.
- No fine-tuning data generated from this flow.

CAREERLY-002 — Home Page Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Home Page — Job Feed & Recommendations
Flow ID:       CAREERLY-002
Trigger:       Authenticated user lands on the home page
Entry Point:   Home screen (post-login or post-setup)
Exit Point:    User views a job detail, or remains browsing the feed
Related Flows: CAREERLY-001 (Auth), CAREERLY-003 (Jobs), CAREERLY-004 (Notifications)
```

1.2 Description

The home page is the core discovery surface of Careerly. It serves two sections: a personalized recommendations section (driven by the user's resume and preferences) and a general job listing section (all scraped jobs, paginated). Jobs are scraped every 15 minutes via a Celery Beat task that pulls from Indeed, LinkedIn, Glassdoor, and Naukri. Each job in the feed is tagged as "new" if it was added in the last 2 hours, or "viewed" if the user has previously opened that job's detail page. The scope of this flow covers the scraping pipeline, the recommendation engine trigger, and the home page rendering — it does not cover the job detail view, which is CAREERLY-003.

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Job Seeker	Human	Views home feed, scrolls recommendations and listings
System	Automated	Serves recommendations and job listing, applies "viewed"/"new" tags
Scraper	Automated	Periodically fetches jobs from external platforms
Celery Beat	Automated	Triggers the scraper task every 15 minutes
AI/ML Service	Automated	Generates job recommendations based on resume + preferences

1.4 Step-by-Step Bullet Points

Sub-flow A — Scraper Pipeline (background, runs every 15 minutes)

- Celery Beat — triggers `scrape_all_platforms` task every 15 minutes
- Celery Task — runs scrapers for Indeed, LinkedIn, Glassdoor, and Naukri in parallel (as a Celery group)
- Scraper (per platform) — fetches latest job listings, normalizes data into the common job schema
  ↳ if platform is unreachable or returns an error: logs the failure, marks that platform's scrape as `FAILED` for this cycle, continues with other platforms
  ↳ if partial data (no description, no skills): saves what is available, marks missing fields as `null` — does not discard the job
- System — for each scraped job, checks if it already exists in the DB (by `external_id` + `platform`)
  ↳ if duplicate: skips insertion, updates `last_seen_at` timestamp on the existing record
  ↳ if new: inserts the job with `status=ACTIVE` , `scraped_at=now`
- System — after all platform scrapers complete, dispatches a notification check:
  - identifies users whose preferences or resume keywords match newly added jobs
  - dispatches Django Task per matched user to send a "new jobs available" notification
- System — logs scrape cycle results (per platform: jobs found, new inserted, duplicates skipped, errors)

Sub-flow B — Home Page Load (user-triggered)

- Job Seeker — lands on the home page (authenticated)
- System — checks if the user has a parsed resume and preferences
  ↳ if resume parsing is still in progress: shows a loading state in the recommendations section with message "Setting up your recommendations..."
  ↳ if resume parsing failed: shows fallback recommendations based on preferences only
- System — checks Redis cache for this user's recommendations (`recommendations:user:{id}`)
  ↳ if cache hit: returns cached recommendations immediately
  ↳ if cache miss: triggers recommendation generation (see Sub-flow C), returns loading state until ready
- System — returns the first page of the general job listing (paginated, 20 per page, sorted by `scraped_at` descending)
- System — for each job in both sections, checks the user's `JobView` history:
  ↳ if job is in the user's viewed history: tags it as `viewed`
  ↳ if job was added in the last 2 hours AND not viewed: tags it as `new`
  ↳ otherwise: no tag
- Job Seeker — sees the recommendations section and the job listing section
- Job Seeker — can scroll and paginate the job listing (infinite scroll or load more)
- Job Seeker — can click any job to go to the job detail (CAREERLY-003)

Sub-flow C — Recommendation Generation (async, triggered on cache miss)

- System — fetches user's parsed resume data (skills, titles, experience) and preferences
- System — queries the jobs DB for jobs matching the user's profile using a relevance ranking query
- AI/ML Service — scores and ranks matched jobs by suitability (uses resume embedding vs job description embedding)
  ↳ if AI service unavailable: falls back to keyword-based matching from preferences only
- System — takes top N recommendations (configurable, default 10)
- System — stores recommendations in Redis cache with TTL of 30 minutes (`recommendations:user:{id}`)
- System — returns recommendations to the home page response

1.5 Validations

Input Validations

Field	Rule	Error Message
Page number (pagination)	Positive integer, min 1	"Invalid page number"
Page size	Max 50 per request	Capped at 50 silently

Business Rule Validations

Rule	Condition	Behavior
Auth required	Unauthenticated user accesses home	Redirect to login
Setup incomplete	User status is not ACTIVE	Redirect to setup screen
Resume not yet parsed	Parsing still in progress	Show loading state in recommendations, serve feed normally
Scrape deduplication	Job with same external_id + platform exists	Skip insert, update last_seen_at only
New tag window	Job scraped_at > now - 2 hours AND not viewed by user	Tag as "new"
Viewed tag	Job exists in user's JobView records	Tag as "viewed" — overrides "new" if both apply

Security Validations

Check	Details
Authentication	JWT required — all home page endpoints are protected
Role-based access	Only Job Seekers access the home feed — admins have separate dashboard
Scraper endpoints	Internal only — not exposed via API, triggered by Celery only

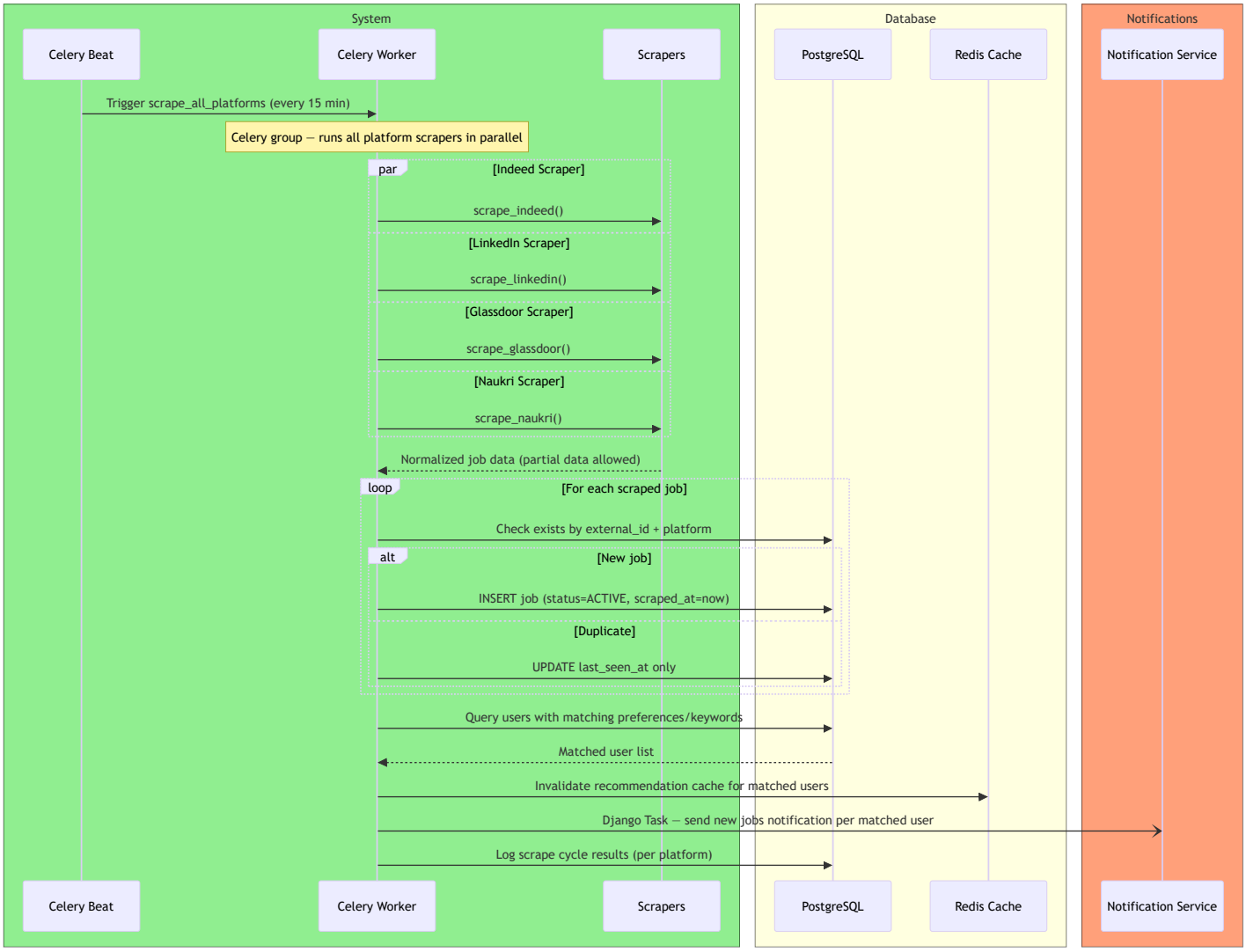
Error Handling

Scenario	System Response
All scrapers fail in a cycle	Log error, alert admin via system notification, serve existing jobs from DB
One platform scraper fails	Continue with others, log failure, show in monitoring dashboard
Recommendations unavailable	Fall back to preference-based listing, no error shown to user
DB query timeout on job listing	Show "Something went wrong. Pull to refresh."
Redis cache unavailable	Fall back to DB query for recommendations, log the cache miss
Empty job listing (no jobs in DB yet)	Show "No jobs yet. Check back soon." empty state

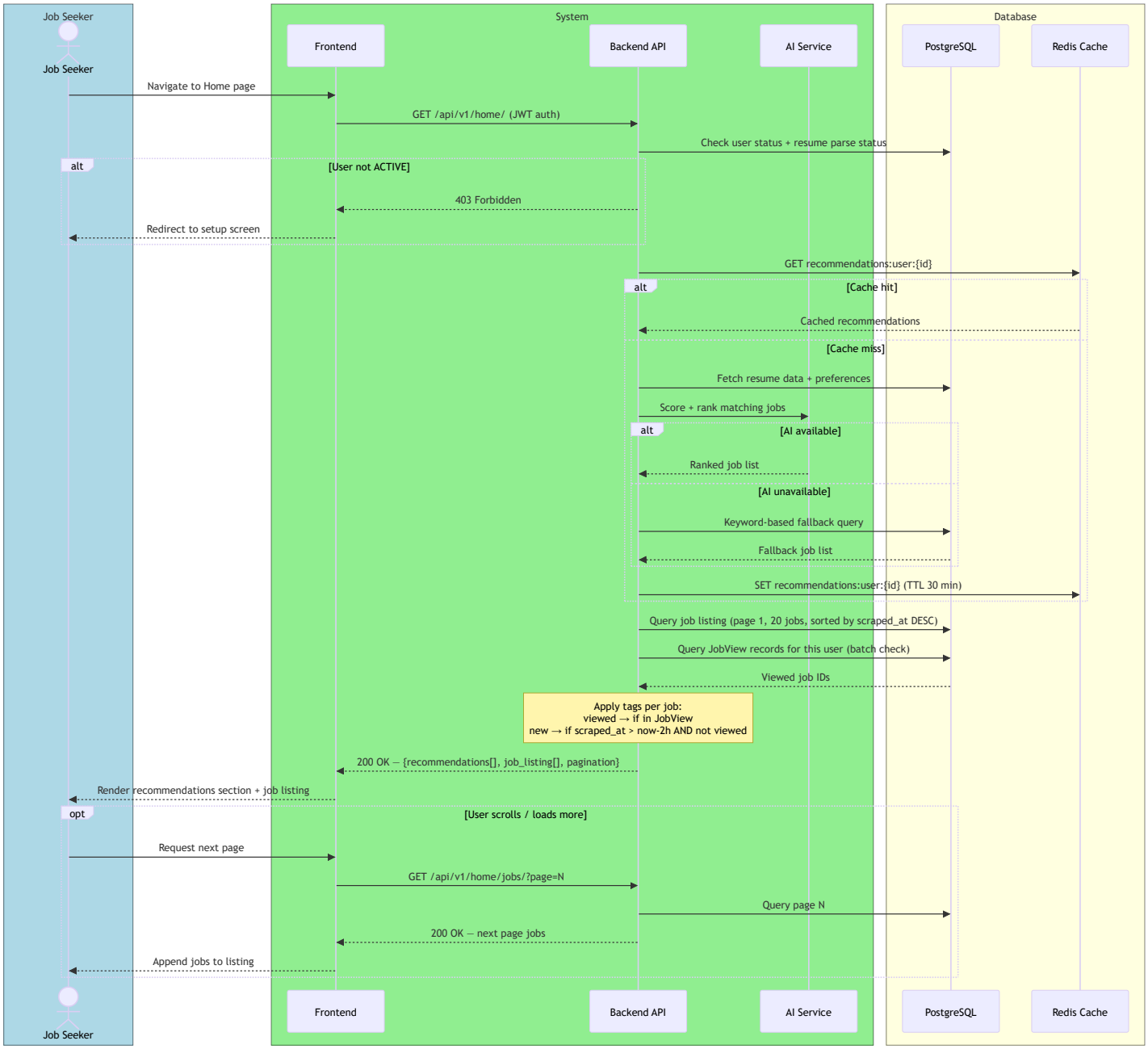
PART 2 — TECHNICAL

2.1 Diagrams

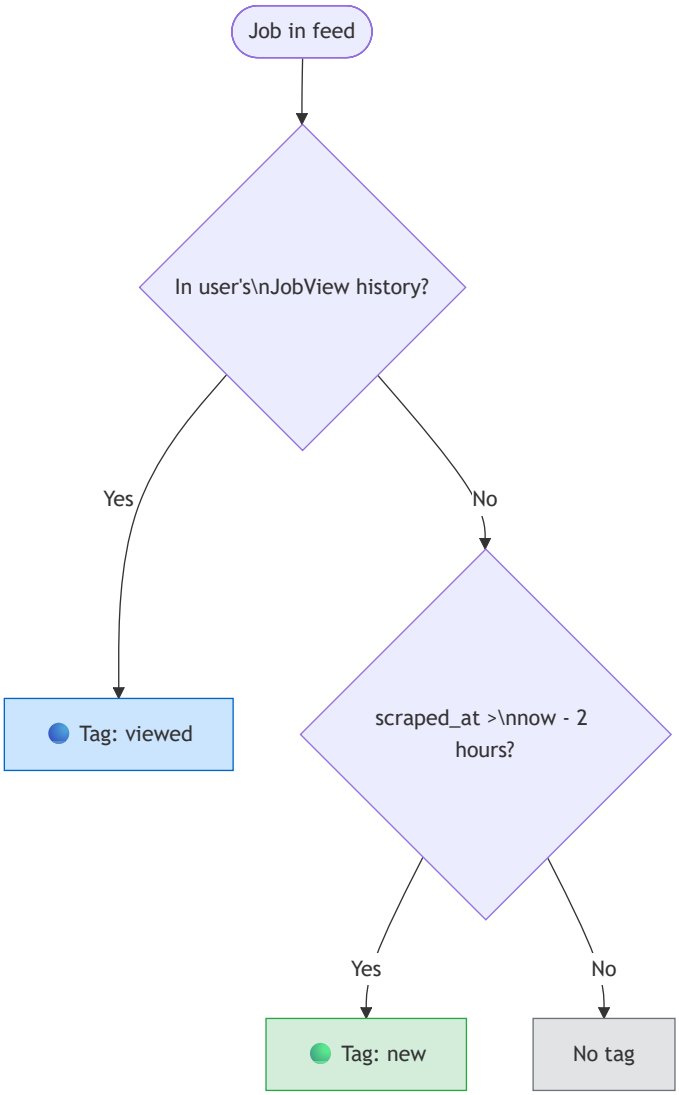
Sequence Diagram — Scraper Pipeline



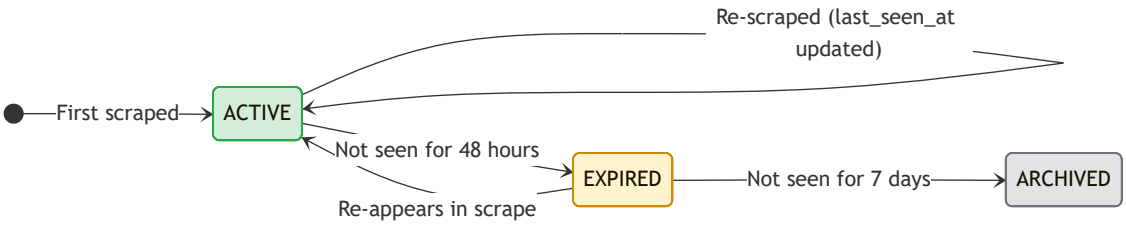
Sequence Diagram — Home Page Load



Flowchart — Job Tag Logic



State Diagram — Scrape Job Status



2.2 Data Models

Model: `Company`

Purpose: Represents a company extracted from scraped job data — shared across multiple job listings to avoid duplication
Django app: `jobs`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`name`	`CharField(max_length=255)`	Yes	—	Company name. Indexed.

Field	Django Field Type	Required	Default	Notes
`industry`	`CharField(max_length=255, null=True, blank=True)`	No	`null`	e.g. Technology, Healthcare
`url`	`URLField(null=True, blank=True)`	No	`null`	Company website
`url_direct`	`URLField(null=True, blank=True)`	No	`null`	Direct company page on the platform
`logo`	`URLField(null=True, blank=True)`	No	`null`	Logo image URL from the platform
`addresses`	`TextField(null=True, blank=True)`	No	`null`	Raw address string(s) as returned by scraper
`num_employees`	`CharField(max_length=50, null=True, blank=True)`	No	`null`	Range string e.g. "1000-5000" — not an integer, platforms return ranges
`revenue`	`CharField(max_length=100, null=True, blank=True)`	No	`null`	Revenue range string as returned by platform
`description`	`TextField(null=True, blank=True)`	No	`null`	Company description
`rating`	`DecimalField(max_digits=3, decimal_places=1, null=True, blank=True)`	No	`null`	Employer rating e.g. 4.2
`reviews_count`	`PositiveIntegerField(null=True, blank=True)`	No	`null`	Number of reviews
`updated_at`	`DateTimeField(auto_now=True)`	Auto	`now`	Updated whenever new scrape brings fresher company data

> **Deduplication note:** Companies are matched by `name` (case-insensitive). If a company appears across multiple platforms, they share one record. Company fields are updated (not overwritten blindly) — only update a field if the incoming value is non-null and the existing value is null, to progressively enrich the record.

Model: `Job`

Purpose: Represents a scraped job listing from any supported platform
Django app: `jobs`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`external_id`	`CharField(max_length=255)`	Yes	—	Job ID from the source platform. Indexed.
`platform`	`CharField(choices=PLATFORM, max_length=20)`	Yes	—	Enum: INDEED, LINKEDIN, GLASSDOOR, NAUKRI
`company`	`ForeignKey(Company, on_delete=SET_NULL, null=True)`	No	`null`	FK to Company — SET_NULL if company record deleted
`title`	`CharField(max_length=255)`	Yes	—	Job title. Indexed for search.
`location`	`CharField(max_length=255, null=True, blank=True)`	No	`null`	City/region — null for fully remote roles
`date_posted`	`DateField(null=True, blank=True)`	No	`null`	Actual posting date from the platform —

Field	Django Field Type	Required	Default	Notes
				different from scraped_at
job_type	CharField(choices=JOB_TYPE, max_length=20, null=True, blank=True)	No	null	Enum: FULL_TIME, PART_TIME, CONTRACT, INTERNSHIP, null if not provided
job_level	CharField(max_length=100, null=True, blank=True)	No	null	Seniority level e.g. "Entry Level", "Senior", "Director"
job_function	CharField(max_length=100, null=True, blank=True)	No	null	Department/function e.g. "Engineering", "Marketing"
description	TextField(null=True, blank=True)	No	null	Full job description — not provided by all platforms
skills	ArrayField(CharField(max_length=100), null=True, blank=True)	No	null	Required skills — not provided by all platforms
experience_range	CharField(max_length=100, null=True, blank=True)	No	null	e.g. "2-5 years" — raw string as returned by scraper
salary_source	CharField(max_length=50, null=True, blank=True)	No	null	Source of salary info e.g. "employer", "estimated"
interval	CharField(max_length=20, null=True, blank=True)	No	null	Pay interval: "yearly", "monthly", "hourly"
min_amount	DecimalField(max_digits=12, decimal_places=2, null=True, blank=True)	No	null	Minimum salary amount
max_amount	DecimalField(max_digits=12, decimal_places=2, null=True, blank=True)	No	null	Maximum salary amount
currency	CharField(max_length=10, null=True, blank=True)	No	null	e.g. "USD", "EGP"
is_remote	BooleanField(null=True)	No	null	True/False/null — null means not specified
work_from_home_type	CharField(max_length=50, null=True, blank=True)	No	null	e.g. "hybrid", "fully_remote", "on_site"
listing_type	CharField(max_length=50, null=True, blank=True)	No	null	e.g. "organic", "sponsored"
vacancy_count	PositiveIntegerField(null=True, blank=True)	No	null	Number of openings — not always provided
emails	ArrayField(EmailField(), null=True, blank=True)	No	null	Contact emails if provided by the platform
job_url	URLField(max_length=1000)	Yes	—	Listing URL on the platform (job_url from scraper)

Field	Django Field Type	Required	Default	Notes
`job_url_direct`	`URLField(max_length=1000, null=True, blank=True)`	No	`null`	Direct application URL if different from listing URL
`search_country`	`CharField(max_length=100, null=True, blank=True)`	No	`null`	The country context used in the scrape search
`search_location`	`CharField(max_length=255, null=True, blank=True)`	No	`null`	The location query used in the scrape search
`status`	`CharField(choices=JOB_STATUS, max_length=20)`	Yes	`ACTIVE`	Enum: ACTIVE, EXPIRED, ARCHIVED
`scraped_at`	`DateTimeField`	Yes	`now`	When first scraped into the system. Indexed.
`last_seen_at`	`DateTimeField`	Yes	`now`	Updated every scrape cycle when job reappears. Indexed.

Unique constraint: `unique_together = [('external_id', 'platform')]`

> **Null handling rule:** Never substitute an empty string for a null. If the scraper returns no value for a field, store `null`. This is enforced on all nullable fields with `null=True, blank=True`.

Model: `ScrapeLog`

Purpose: Logs each scrape cycle per platform for monitoring and debugging
Django app: `jobs`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`platform`	`CharField(choices=PLATFORM, max_length=20)`	Yes	—	Which platform this log is for
`status`	`CharField(choices=SCRAPE_STATUS, max_length=20)`	Yes	—	Enum: SUCCESS, PARTIAL, FAILED
`jobs_found`	`PositiveIntegerField`	No	`0`	Total jobs returned by platform
`jobs_inserted`	`PositiveIntegerField`	No	`0`	New jobs inserted this cycle
`jobs_updated`	`PositiveIntegerField`	No	`0`	Existing jobs with updated last_seen_at
`error_message`	`TextField(null=True, blank=True)`	No	`null`	Error details if status = FAILED
`started_at`	`DateTimeField`	Yes	—	When this platform's scraper started
`finished_at`	`DateTimeField(null=True)`	No	`null`	When it finished (null if still running)

Model: `JobView`

Purpose: Tracks which jobs a user has viewed — used for "viewed" tags and profile stats
Django app: `jobs`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	The viewing user
`job`	`ForeignKey(Job, on_delete=CASCADE)`	Yes	—	The viewed job

Field	Django Field Type	Required	Default	Notes
`viewed_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	Indexed — used for weekly stats

Unique constraint: `unique_together = [('user', 'job')]` — one record per user-job pair. Update `viewed_at` on revisit rather than inserting a new record.

Model: `SavedJob`

Purpose: Jobs saved by a user for later reference
Django app: `jobs`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	The saving user
`job`	`ForeignKey(Job, on_delete=CASCADE)`	Yes	—	The saved job
`saved_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Unique constraint: `unique_together = [('user', 'job')]`

2.3 Table Relationships & Logic

`Company` and `Job` have a `ForeignKey` relationship — one company has many jobs. `Job` uses `SET_NULL` on company deletion so job records are never lost if a company record is cleaned up. `JobView` and `SavedJob` are junction tables between `User` and `Job`. When a `User` is deleted, all their `JobView` and `SavedJob` records cascade-delete. When a `Job` is deleted, the same applies.

Company deduplication — companies are matched by name (case-insensitive) using `iexact` lookup before creating a new record:

```
company, created = Company.objects.get_or_create(
    name__iexact=scraped_name,
    defaults={'name': scraped_name, ...}
)
```

If the company already exists, enrich it — only update null fields with non-null incoming values. Never overwrite existing data with null:

```
for field in ['logo', 'url', 'description', 'rating', ...]:
    incoming = scraped_data.get(field)
    if incoming and not getattr(company, field):
        setattr(company, field, incoming)
company.save()
```

Job deduplication uses `update_or_create` on `(external_id, platform)`. If the record exists, update `last_seen_at` only — do not overwrite any other fields. This preserves the original scraped data integrity.

Tag computation is done in the backend, not the frontend. When building the home page response, the backend fetches the user's `JobView` IDs as a set, then annotates each job in the response:

```
tag = "viewed" if job.id in viewed_ids
     else "new" if job.scraped_at > now - 2h
     else None
```

This is computed in Python after the DB query — do not compute tags in SQL to keep the query simple. The computed `tag` value is included as a field on every job object in the API response (`"new"` , `"viewed"` , or `null`). The frontend reads `job.tag` directly — no tag logic lives on the client.

Job expiry — a separate Celery Beat task runs daily: marks jobs as `EXPIRED` if `last_seen_at < now - 48 hours`. Marks `EXPIRED` jobs as `ARCHIVED` if `last_seen_at < now - 7 days`. Expired/archived jobs are excluded from the home feed by default (filter `status=ACTIVE`).

Recommendation cache invalidation — when new jobs matching a user are inserted, invalidate their recommendations cache key in Redis. This ensures the next home page load recalculates recommendations with fresh data.

Scraper field mapping — the scraper output fields map to models as follows:

Scraper Field	Model	Model Field
id	Job	external_id
site	Job	platform
job_url	Job	job_url
job_url_direct	Job	job_url_direct
title	Job	title
company	Company	name
location	Job	location
date_posted	Job	date_posted
job_type	Job	job_type
salary_source	Job	salary_source
interval	Job	interval
min_amount	Job	min_amount
max_amount	Job	max_amount
currency	Job	currency
is_remote	Job	is_remote
job_level	Job	job_level
job_function	Job	job_function
listing_type	Job	listing_type
emails	Job	emails
description	Job	description
company_industry	Company	industry
company_url	Company	url
company_logo	Company	logo
company_url_direct	Company	url_direct
company_addresses	Company	addresses
company_num_employees	Company	num_employees
company_revenue	Company	revenue
company_description	Company	description
skills	Job	skills
experience_range	Job	experience_range
company_rating	Company	rating
company_reviews_count	Company	reviews_count
vacancy_count	Job	vacancy_count
work_from_home_type	Job	work_from_home_type

Scraper Field	Model	Model Field
`search_country`	`Job`	`search_country`
`search_location`	`Job`	`search_location`
(set by system)	`Job`	`scraped_at`
(set by system)	`Job`	`last_seen_at`

Indexes needed:

- `Job.scraped_at` — used for sorting the feed
- `Job.last_seen_at` — used for expiry checks
- `Job.title` — used for search and recommendation matching
- `Job.external_id` + `Job.platform` — covered by unique constraint
- `Company.name` — used for deduplication lookup
- `JobView.viewed_at` — used for weekly stats in monitoring
- `JobView.user` — used for tag lookups

2.4 API Endpoints

Method	Endpoint	Auth	Role	Request Body / Params	Response	Des
`GET`	`/api/v1/home/`	Yes	Job Seeker	—	`200` — `{recommendations[], job_listing[], pagination}`	Home — recom + first listings
`GET`	`/api/v1/home/jobs/`	Yes	Job Seeker	`?page=N&page_size=20`	`200` — `{jobs[], pagination}`	Pagina listing
`GET`	`/api/v1/home/recommendations/`	Yes	Job Seeker	—	`200` — `{jobs[]}`	Recom only (f

2.5 Developer Notes

🔵 Backend Developer (Django)

- Celery Beat schedule: `scrape_all_platforms` every 15 minutes via `crontab(minute='*/15')`.
- Use `celery.group` to run all 4 platform scrapers in parallel within one Celery task. Each scraper is a separate `shared_task`.
- Scrapers should be stateless — no shared mutable state between runs. Each scraper returns a list of normalized job dicts that map directly to the scraper field mapping table above.
- **Company upsert first, then Job upsert.** For each scraped job: (1) `get_or_create` the Company by `name__iexact`, enrich null fields. (2) `update_or_create` the Job by `(external_id, platform)`, set `company=company_instance`, only update `last_seen_at` on existing records.
- Never overwrite existing non-null Job fields on update — only `last_seen_at` is updated on re-scrape. This protects data integrity if the platform returns degraded data in a later cycle.
- All nullable fields must be stored as `null`, never as empty string. Apply this strictly in the scraper normalization layer before hitting the DB.
- `is_remote` is a `NullBooleanField` — three states: `True`, `False`, `None` (not specified). Do not default to `False`.
- `skills` and `emails` are `ArrayField` — use `[]` as the default in Python but store `null` in DB when not provided (use `null=True`).
- Recommendation endpoint: use `select_related('company')` when fetching jobs — avoids N+1 on company name/logo lookups.
- `JobView` upsert: `update_or_create(user=user, job=job, defaults={'viewed_at': now()})`.
- Add a daily Celery Beat task `expire_old_jobs`: `Job.objects.filter(last_seen_at__lt=now()-48h, status='ACTIVE').update(status='EXPIRED')`.
- For the notification dispatch after scraping: collect new job IDs, find users with matching `job_titles` or `skills` overlap, batch them, dispatch one Django Task per user.
- Use `select_related('company')` and `only()` for the job listing query to avoid pulling unnecessary fields.

- The job serializer must include a `tag` field as a `SerializerMethodField` — compute it from the pre-fetched `viewed_ids` set passed into the serializer context. Never make a per-job DB query to determine the tag.

🟢 Frontend Developer (React)

- Home page has two distinct sections: **Recommendations** (horizontal scroll or grid, top) and **Job Listing** (vertical list, below).
- Each job card displays: title, company name, company logo (if available — fallback to initials avatar), location, platform badge, job type, remote badge (if `is_remote=true`), "new" or "viewed" tag (colored), `date_posted` (preferred over `scraped_at` for display).
- Recommendations section shows a skeleton loader while loading. If recommendations are not ready (resume parsing in progress), show a banner: "We're personalizing your feed — check back shortly."
- Job listing uses **infinite scroll** — trigger `GET /api/v1/home/jobs/?page=N` when user scrolls to 80% of the list. Append results to the existing list.
- Tag rendering: read `job.tag` from the API response directly — `"new"` → green badge, `"viewed"` → grey badge, `null` → nothing. Never compute tags on the frontend.
- Platform badge: show platform logo/icon (Indeed, LinkedIn, Glassdoor, Naukri).
- Salary display: show only if `min_amount` or `max_amount` is non-null. Format as `$X – $Y /year` using `interval` and `currency`. If only one bound is available, show `From $X` or `Up to $Y`.
- Clicking a job card navigates to the job detail page (CAREERLY-003) and fires a `POST /api/v1/jobs/{id}/view/` to record the view.
- Cache the home page data in React Query or SWR with a 5-minute stale time — don't re-fetch on every re-render.

🟡 Mobile Developer (Flutter)

- Recommendations section: `ListView` with `scrollDirection: Axis.horizontal`.
- Job listing: `ListView.builder` with `LazyLoading` — load next page when user reaches last 5 items.
- Job cards: use `Card` widget with platform icon, title, company, location, tag badge.
- Tag badges: `Chip` widget — read `job.tag` from the API response. Green `Chip` for `"new"`, grey for `"viewed"`, nothing rendered for `null`. No tag logic on mobile.
- Pull-to-refresh: wrap listing in `RefreshIndicator` — re-calls `/api/v1/home/` on pull.
- If recommendations are loading (parse in progress): show `Shimmer` loading effect in the recommendations row.
- Store first page of home data in local cache (using `hive` or `isar`) for offline viewing of previously loaded jobs.
- Platform icons: bundle as local assets — do not fetch from the web at runtime.

🟣 AI Engineer

- Recommendation generation: given a user's resume data (skills list, job titles, experience level), query the `Job` table for `ACTIVE` jobs. Score each by relevance using an embedding similarity model.
- Preferred approach: pre-compute job embeddings and store them (as a `vector` field using `pgvector` extension on PostgreSQL, or as a separate vector store). On recommendation request, compute the user's profile embedding and run a nearest-neighbor search.
- If `pgvector` is not available at this stage: fall back to PostgreSQL full-text search using `SearchVector` on `title + description + skills`.
- Input to recommendation model: `{user_skills: [], user_titles: [], experience_level: str}`.
- Output: list of `job_id` ranked by suitability score.
- TTL of recommendations is 30 minutes — cache is invalidated when new matching jobs arrive.
- This same recommendation engine is used in AI Match (CAREERLY-003) — share the scoring logic.

2.6 General Notes

Null field UI behavior: Fields critical to the job card and detail page (`description`, `skills`, `salary`) should show styled placeholders directing the user to the original post — e.g. *"Description not available for this listing. View the original post for full details."* Minor supplementary fields (`vacancy_count`, `listing_type`, `emails`, `work_from_home_type`) should be silently omitted if null — do not render the label at all. The general rule: if the user would notice it's missing, show a placeholder; if it's supplementary, hide the row entirely. Decision needed: confirm placeholder copy with design before the frontend builds the job detail component.

CAREERLY-003 — Jobs Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Jobs — Detail, Save, Apply, AI Match, Check Resume
Flow ID:       CAREERLY-003
Trigger:       User clicks on a job card from the home page or job listing
Entry Point:   Job card on Home page or Job listing
Exit Point:
  Apply Now    → External job page (browser/webview)
  AI Match     → AI analysis result screen with suitability score + top 3 jobs
  Check Resume → AI ATS analysis result screen with ATS score
  Save         → Job saved, user stays on detail page
Related Flows: CAREERLY-002 (Home Page), CAREERLY-004 (Notifications), CAREERLY-005 (Profile)
```

1.2 Description

This flow covers everything that happens after a user selects a job. The job detail page presents scraped and organized data about the job, and offers four actions: save for later, apply now (external redirect), AI Match (resume vs job suitability scoring), and Check Your Resume (general ATS scoring). Both AI features are analysis sessions — they persist to the user's history and contribute to profile stats. The AI Match session is scoped to a specific job; the Check Resume session is general. Both support uploading additional resumes for comparison within the same session, and both render AI responses in markdown. This flow does not cover the scraping of jobs (CAREERLY-002) or notifications triggered by these actions (CAREERLY-004).

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Job Seeker	Human	Views job detail, saves, applies, requests AI analysis
System	Automated	Records views, saves, routes to external URL, manages sessions
AI/ML Service	Automated	Analyzes resume against job or general ATS, returns scored markdown report

1.4 Step-by-Step Bullet Points

Route 1 — View Job Detail

- Job Seeker — clicks on a job card
- System — records a `JobView` for this user + job (upsert — update `viewed_at` if already exists)
- System — fetches job details from DB
  ↳ if job no longer exists or is archived: shows "This job is no longer available"
- Job Seeker — sees the job detail page with all available scraped fields (title, company, location, description, skills, job type, salary if available, platform badge, source link)
- System — shows a notice for fields that were not available from the source platform (e.g. "Description not available for this listing")

Route 2 — Save Job

- Job Seeker — clicks the Save button on the detail page
- System — checks if the job is already saved by this user
  ↳ if already saved: toggles to unsave — removes from saved jobs, button state changes to "Save"
  ↳ if not saved: saves the job, button state changes to "Saved"
- System — creates or deletes a `SavedJob` record

- Job Seeker — stays on the job detail page, button reflects current state

Route 3 — Apply Now

- Job Seeker — clicks the "Apply Now" button
- System — retrieves the `external_url` for this job
- System — opens the external job page (opens in a new tab on web, in-app browser or external browser on mobile)
- Job Seeker — lands on the original job page (Indeed, LinkedIn, Glassdoor, or Naukri) to complete the application there

Route 4 — AI Match

- Job Seeker — clicks the "AI Match" button on the job detail page
- System — checks if the user has a resume on file
  ↳ if no resume: prompts user to upload one — "You need a resume to use AI Match"
- System — creates a new `AISession` record with type = `JOB_MATCH`, linked to the job, status = `PENDING`
- Job Seeker — is navigated to the AI Match screen, which shows the job title and company at the top
- System — dispatches Celery Task to analyze the user's default resume against this job
- System — streams or polls analysis progress
- AI/ML Service — analyzes the resume against the job description and requirements
- AI/ML Service — returns a markdown-formatted report including:
  - Suitability score (0–100)
  - Verdict tier (Poor / Fair / Good / Excellent)
  - Strengths and gaps breakdown
  - Top 3 recommended job titles based on the resume
- System — resolves the top 3 recommended job titles against the jobs DB (best match per title)
- System — updates `AISession` with: `score`, `verdict`, `report_markdown`, `top_jobs[]`, `status = COMPLETE`
- Job Seeker — sees the rendered markdown report, suitability score, verdict badge, and the 3 recommended jobs
- Job Seeker — can click any of the 3 recommended jobs to view their detail (re-enters this flow)
- Job Seeker — can upload a second resume to compare against the same job
  ↳ System — creates a new `AISessionResume` record linked to this session
  ↳ System — runs the same AI analysis for the new resume against the same job
  ↳ Job Seeker — sees both results side-by-side or as separate tabs

Route 5 — Check Your Resume (ATS Analysis)

- Job Seeker — clicks the "Check Your Resume" button (available on the job detail page and as a standalone feature)
- System — checks if the user has a resume on file
  ↳ if no resume: prompts user to upload one
- System — creates a new `AISession` record with type = `ATS_CHECK`, status = `PENDING`
- Job Seeker — is navigated to the Check Resume screen
- System — dispatches Celery Task to run a general ATS analysis on the user's default resume
- AI/ML Service — analyzes the resume for ATS compliance and returns a markdown-formatted report including:
  - ATS score (0–100)
  - Verdict tier (Poor / Fair / Good / Excellent)
  - Keyword coverage analysis
  - Formatting recommendations
  - Skills gap suggestions
- System — updates `AISession` with: `score`, `verdict`, `report_markdown`, `status = COMPLETE`
- Job Seeker — sees the rendered markdown report, ATS score, and verdict badge
- Job Seeker — can upload a second resume to compare ATS scores
  ↳ System — runs the same ATS analysis for the new resume
  ↳ Job Seeker — sees both results side-by-side or as separate tabs

1.5 Validations

Input Validations

Field	Rule	Error Message
Resume upload (AI features)	PDF only, max 5MB	"Only PDF files are accepted" / "File size must be under 5MB"
Job ID (URL param)	Must be a valid UUID, must exist in DB	404 Not Found

Business Rule Validations

Rule	Condition	Behavior
Resume required for AI features	No resume on file	Prompt upload — block AI feature until resumed
Archived/expired job viewed	Job status is not ACTIVE	Show "This job is no longer available" — disable Apply and AI Match
Save toggle	Job already saved	Toggle to unsaved on second click
Top 3 job resolution	AI returns a title with no DB match	Return the closest available match; if none, omit that slot (can return 1–3 jobs)
Second resume in session	User uploads 2nd resume	Runs AI analysis for the new resume against the same job/context as the first

Security Validations

Check	Details
Authentication	JWT required for all job actions
Role-based access	Only Job Seekers — admins do not interact with job detail
Session ownership	AI sessions can only be viewed by the user who created them
File upload	Validate MIME type server-side (not just extension) — reject non-PDF content types

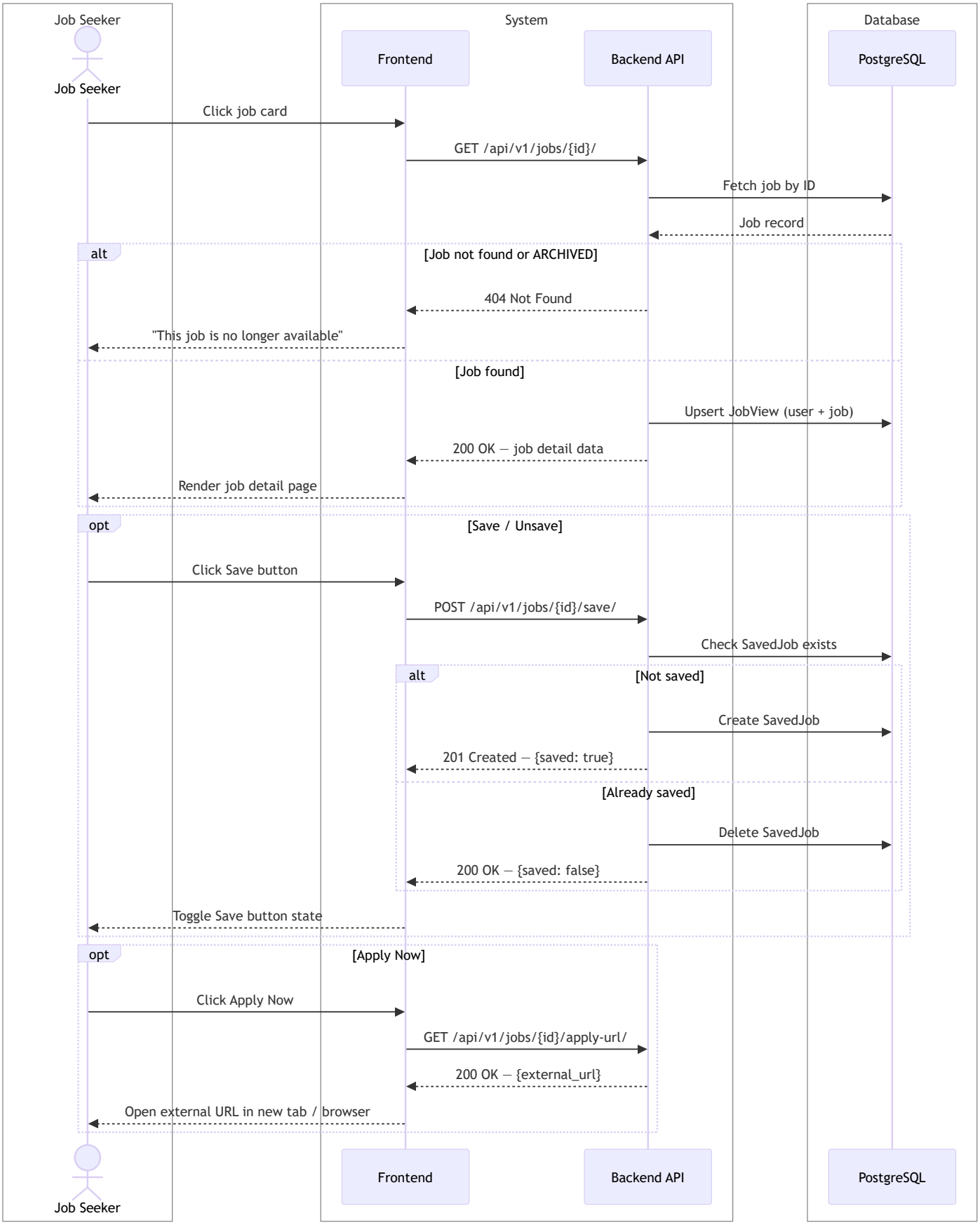
Error Handling

Scenario	System Response
AI service unavailable	Show "AI analysis is temporarily unavailable. Try again later." — do not create a session
AI analysis times out	Mark session status = `FAILED` , notify user — allow retry
External URL missing or broken	Show "Apply link is unavailable for this listing" — disable Apply Now button
Job not found in DB	404 with message "This job is no longer available"
Save/unsave server error	Show toast error, revert button state

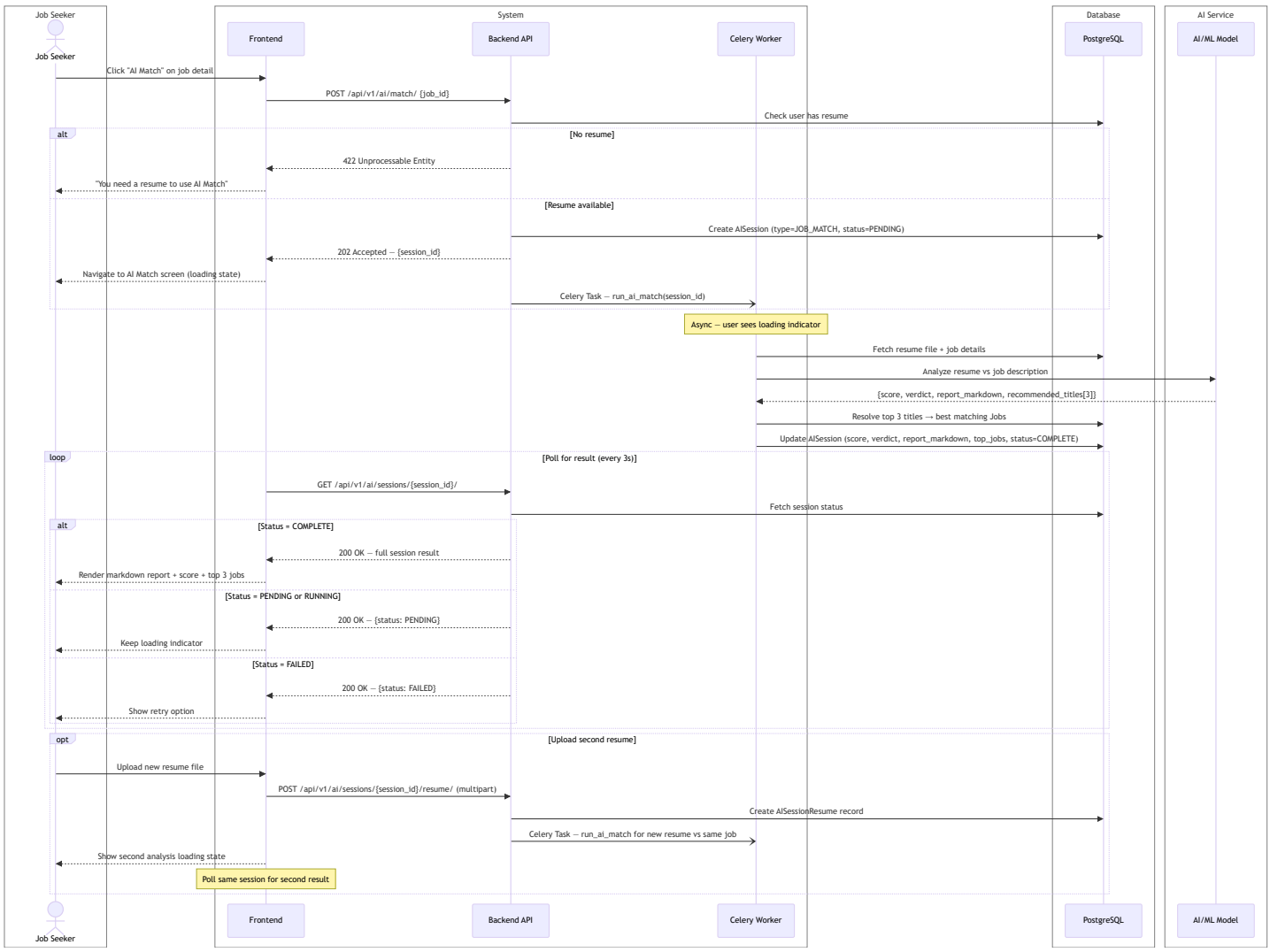
PART 2 — TECHNICAL

2.1 Diagrams

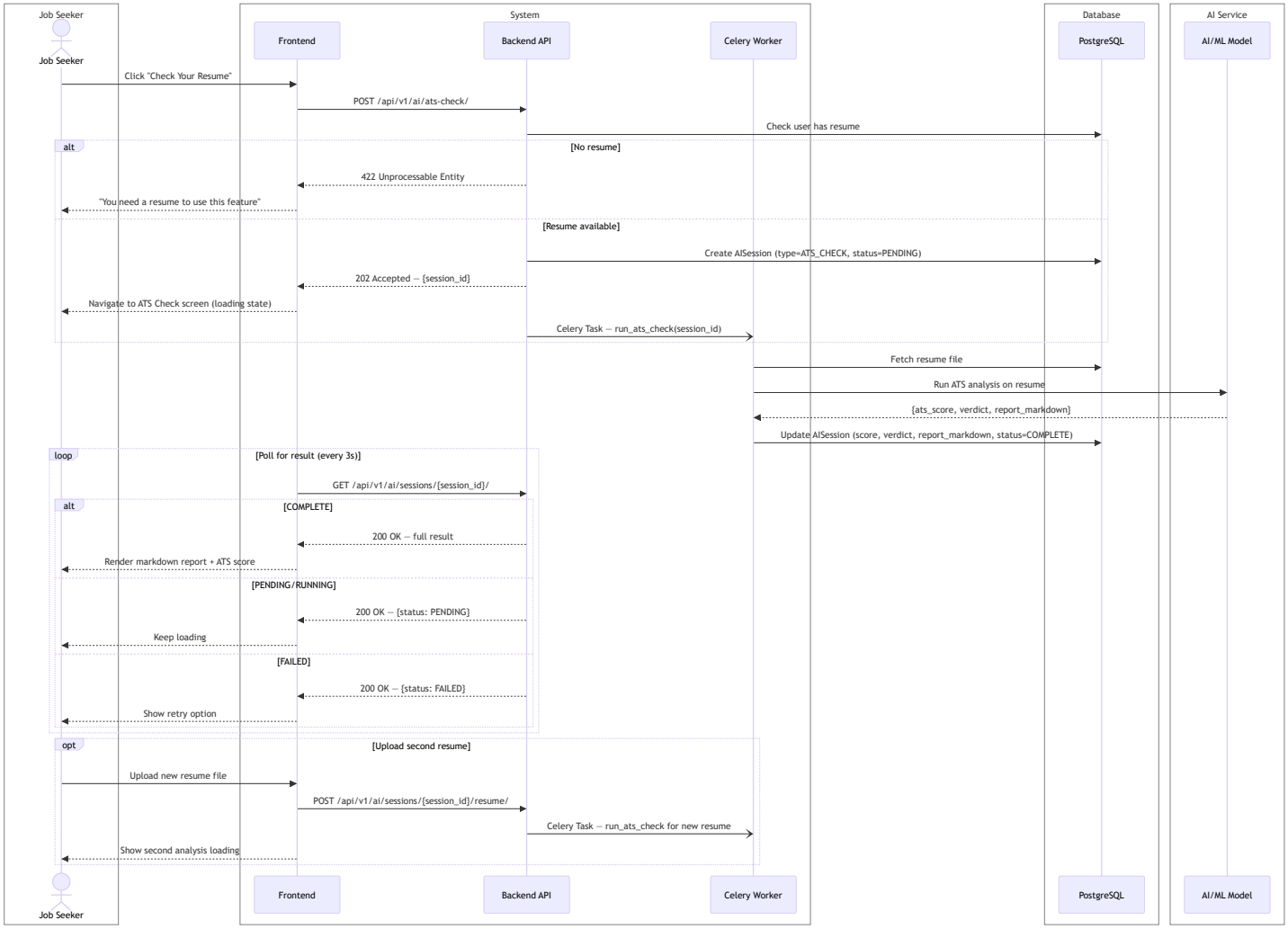
Sequence Diagram — Job Detail + Save + Apply


Job Seeker
System
Database
Job Seeker
Frontend
Backend API
PostgreSQL
Click job card
GET /api/v1/jobs/{id}/
Fetch job by ID
Job record
alt
[Job not found or ARCHIVED]
404 Not Found
"This job is no longer available"
[Job found]
Upsert JobView (user + job)
200 OK — job detail data
Render job detail page
opt
[Save / Unsave]
Click Save button
POST /api/v1/jobs/{id}/save/
Check SavedJob exists
alt
[Not saved]
Create SavedJob
201 Created — {saved: true}
[Already saved]
Delete SavedJob
200 OK — {saved: false}
Toggle Save button state
opt
[Apply Now]
Click Apply Now
GET /api/v1/jobs/{id}/apply-url/
200 OK — {external_url}
Open external URL in new tab / browser
Job Seeker
Frontend
Backend API
PostgreSQL

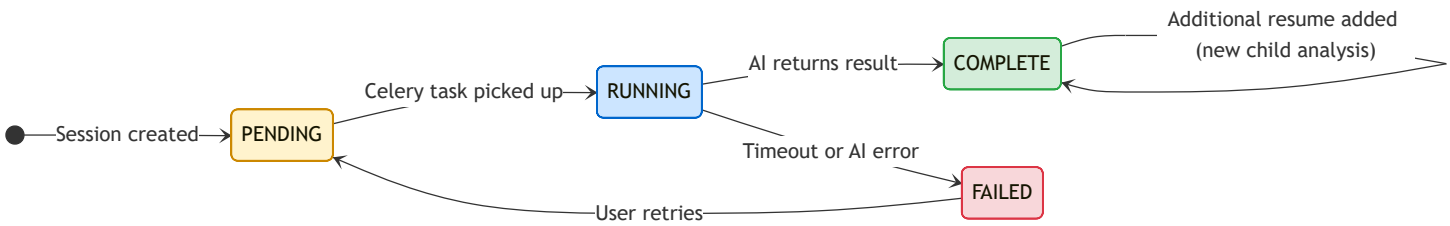
Sequence Diagram — AI Match



Sequence Diagram — Check Your Resume (ATS)



State Diagram — AI Session Lifecycle



2.2 Data Models

Model: `AISession`

Purpose: Represents one AI analysis session — either a job match or ATS check

Django app: `ai`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Owning user
`session_type`	`CharField(choices=SESSION_TYPE, max_length=20)`	Yes	—	Enum: JOB_MATCH, ATS_CHECK

Field	Django Field Type	Required	Default	Notes
`job`	`ForeignKey(Job, on_delete=SET_NULL, null=True, blank=True)`	No	`null`	Only for JOB_MATCH sessions
`resume`	`ForeignKey(Resume, on_delete=SET_NULL, null=True)`	Yes	—	The primary resume analyzed
`score`	`PositiveSmallIntegerField(null=True)`	No	`null`	0–100, set on completion
`verdict`	`CharField(choices=VERDICT, max_length=20, null=True)`	No	`null`	Enum: POOR, FAIR, GOOD, EXCELLENT
`report_markdown`	`TextField(null=True, blank=True)`	No	`null`	Full AI-generated markdown report
`top_jobs`	`ManyToManyField(Job, blank=True, related_name='recommended_in')`	No	—	Top 3 jobs (JOB_MATCH only)
`status`	`CharField(choices=SESSION_STATUS, max_length=20)`	Yes	`PENDING`	Enum: PENDING, RUNNING, COMPLETE, FAILED
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	Indexed — used for session history
`completed_at`	`DateTimeField(null=True, blank=True)`	No	`null`	Set when status = COMPLETE

Model: `AISessionResume`

Purpose: Tracks additional resumes analyzed within the same session (for comparison)
Django app: `ai`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`session`	`ForeignKey(AISession, on_delete=CASCADE)`	Yes	—	Parent session
`resume`	`ForeignKey(Resume, on_delete=CASCADE)`	Yes	—	The additional resume
`score`	`PositiveSmallIntegerField(null=True)`	No	`null`	Score for this resume in this session
`verdict`	`CharField(choices=VERDICT, max_length=20, null=True)`	No	`null`	Verdict for this resume
`report_markdown`	`TextField(null=True, blank=True)`	No	`null`	Separate markdown report for this resume
`status`	`CharField(choices=SESSION_STATUS, max_length=20)`	Yes	`PENDING`	Same enum as AISession
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Model: `Resume`

Purpose: A resume file uploaded by a user
Django app: `accounts`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Owner
`file`	`FileField(upload_to='resumes/')`	Yes	—	Stored in S3 or local media
`original_filename`	`CharField(max_length=255)`	Yes	—	Displayed in UI and sessions
`file_size`	`PositiveIntegerField`	Yes	—	In bytes
`parsed_data`	`JSONField(null=True, blank=True)`	No	`null`	AI-extracted: skills, titles, experience_level, education
`parse_status`	`CharField(choices=PARSE_STATUS, max_length=20)`	Yes	`PENDING`	Enum: PENDING, COMPLETE, FAILED

Field	Django Field Type	Required	Default	Notes
`is_default`	`BooleanField`	No	`False`	The resume used by default for AI features
`uploaded_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Constraint: Only one `Resume` per user can have `is_default=True`. Enforce with a `UniqueConstraint` filtered on `is_default=True`.

2.3 Table Relationships & Logic

`AISession` links `User`, `Job` (optional), and `Resume`. When a `Job` is deleted or archived, the FK is set to `null` — sessions are preserved for historical record. When a `User` is deleted, sessions cascade-delete.

`AISessionResume` is a child of `AISession`. Each additional resume uploaded in a session creates one `AISessionResume` record with its own independent score, verdict, and markdown report.

Verdict tier computation — applied after score is returned by AI:

```
85–100  → EXCELLENT
70–84   → GOOD
50–69   → FAIR
0–49    → POOR
```

This is computed in the backend before saving — the AI returns only a numeric score.

Average ATS score on profile — computed as:

```
AISession.objects.filter(
    user=user,
    session_type='ATS_CHECK',
    status='COMPLETE'
).aggregate(avg=Avg('score'))
```

This is not stored — it is computed on profile load and cached in Redis with TTL 10 minutes.

Average suitability score on profile — same approach, filtered on `session_type='JOB_MATCH'`.

Default resume logic — when a user uploads their first resume, it is automatically set as `is_default=True`. When they upload subsequent resumes, they can explicitly set a new default. When the default resume is deleted, set the most recently uploaded remaining resume as the new default via `post_delete` signal.

Top 3 job resolution — AI returns 3 job title strings. For each title, run:

```
Job.objects.filter(
    status='ACTIVE',
    title__icontains=title
).order_by('-scraped_at').first()
```

If no match, omit that slot. Store the resolved `Job` instances in the `top_jobs` M2M field.

Session polling — frontend polls every 3 seconds. Backend returns current status. Once `COMPLETE`, frontend stops polling and renders the result. Celery task timeout is 5 minutes — if exceeded, mark session as `FAILED`.

2.4 API Endpoints

Method	Endpoint	Auth	Role	Request Body / Params	Response	Description
`GET`	`/api/v1/jobs/{id}/`	Yes	Job Seeker	—	`200` — job detail	Get job details, records view

Method	Endpoint	Auth	Role	Request Body / Params	Response	Description
`POST`	`/api/v1/jobs/{id}/save/`	Yes	Job Seeker	—	`201 / 200` — `{saved: bool}`	Toggle save/unsave
`POST`	`/api/v1/ai/match/`	Yes	Job Seeker	`{job_id}`	`202` — `{session_id}`	Start AI Match session
`POST`	`/api/v1/ai/ats-check/`	Yes	Job Seeker	—	`202` — `{session_id}`	Start ATS Check session
`GET`	`/api/v1/ai/sessions/{id}/`	Yes	Job Seeker	—	`200` — session data	Poll session status and result
`POST`	`/api/v1/ai/sessions/{id}/resume/`	Yes	Job Seeker	`multipart: {resume_file}`	`202` — `{sub_session_id}`	Add additional resume to session
`GET`	`/api/v1/ai/sessions/`	Yes	Job Seeker	`?type=JOB_MATCH|ATS_CHECK&page=N`	`200` — paginated sessions	List user's session history

2.5 Developer Notes

🔵 Backend Developer (Django)

- `GET /api/v1/jobs/{id}/` should upsert `JobView` inside the view using `update_or_create` — do not fire a separate request for view tracking.
- `POST /api/v1/jobs/{id}/save/` should be idempotent — second call toggles the save, not duplicates it. Use `get_or_create` then delete if it existed.
- AI sessions return `202 Accepted` immediately — the Celery task runs async. Never make the user wait synchronously for AI response.
- Celery task `run_ai_match(session_id)` : fetch session → fetch resume file → send to AI → parse response → compute verdict → resolve top 3 jobs → update session. Wrap in `try/except` — on any exception, set `status=FAILED` .
- Set `CELERY_TASK_SOFT_TIME_LIMIT = 280` and `CELERY_TASK_TIME_LIMIT = 300` for AI tasks (5 min hard limit).
- Markdown in `report_markdown` : store exactly as returned by AI — do not sanitize or strip. The frontend handles rendering.
- For `top_jobs` resolution: do the DB query inside the Celery task, not in the API view.
- Average score computation for profile: use Django's `Avg` aggregation, cache in Redis for 10 minutes. Key: `user:{id}:avg_ats_score` and `user:{id}:avg_match_score` .
- Resume `is_default` uniqueness: use `UniqueConstraint(fields=['user'], condition=Q(is_default=True), name='unique_default_resume_per_user')` .
- `post_delete` signal on `Resume` : if deleted resume was the default, find the latest remaining resume and set it as default.

🟢 Frontend Developer (React)

- Job detail page sections: header (title, company, location, platform badge, save button, apply button), body (description with null notice if missing, skills chips or null notice, job type, salary if available), AI actions section (AI Match button, Check Resume button).
- Save button: optimistic UI — toggle state immediately, revert on API error.
- "AI Match" and "Check Resume" buttons should be disabled and show tooltip "Upload a resume first" if user has no resume.
- AI result screen: render `report_markdown` using a markdown renderer (e.g. `react-markdown` with `remark-gfm` plugin for tables and lists). Never render raw HTML.
- Polling: use `setInterval` every 3 seconds while `status === 'PENDING' || status === 'RUNNING'` . Clear interval on `COMPLETE` or `FAILED` . Use React Query's `refetchInterval` option for cleaner implementation.
- Score display: large circular progress indicator showing the score (0–100). Color-coded by verdict: POOR = red, FAIR = orange, GOOD = blue, EXCELLENT = green.
- Verdict badge: pill/chip component with verdict label and matching color.
- Top 3 recommended jobs: horizontal job cards below the report — clickable, navigate to their detail pages.
- Second resume comparison: show two columns (or tabs on mobile) — "Resume 1" and "Resume 2" — each with their own score, verdict, and report.
- Apply Now: always open in a new tab (`target="_blank"` with `rel="noopener noreferrer"`).

🟡 Mobile Developer (Flutter)

- Job detail: `SingleChildScrollView` with sections. Use `Chip` widgets for skills. Show "Not available" `Text` in grey for null fields.
- Apply Now: use `url_launcher` package to open external URL. Try external browser first, fall back to in-app webview.

- Save button: `IconButton` with bookmark icon — filled when saved, outlined when not. Optimistic toggle.
- AI Match / ATS result screen: use `flutter_markdown` package to render `report_markdown`. Ensure the package supports tables and code blocks.
- Score display: use `CircularProgressIndicator` styled as a score ring, with color driven by verdict.
- Polling: use a `Timer.periodic` in the state — cancel it when status is `COMPLETE` or `FAILED`.
- Second resume: use a `TabBar` with two tabs ("Resume 1", "Resume 2") — each tab holds a full result view.
- AI features unavailable (no resume): show a `BottomSheet` prompting resume upload, with a direct upload button.

AI Engineer

AI Match:

- Input: resume text (extracted from PDF), job title, job description (may be null — handle gracefully), required skills (may be null).
- Output (JSON): `{score: int, recommended_titles: [str, str, str], report_markdown: str}`
- If description is null: base analysis on job title + skills only — note this limitation in the report.
- The markdown report must include: an executive summary, a strengths section, a gaps section, and recommendations.
- Use structured prompting with explicit markdown formatting instructions — the model must output markdown, not plain text.

ATS Check:

- Input: resume text only.
- Output (JSON): `{score: int, report_markdown: str}`
- The markdown report must include: ATS compatibility summary, keyword coverage, formatting issues, improvement recommendations.

Verdict tier: backend computes verdict from score — AI does not return a verdict string.

Second resume comparison: AI is called independently for each resume — there is no "compare two resumes" prompt. The frontend handles the side-by-side display.

Fallback: if AI service is unavailable (connection error, timeout), raise an exception in the Celery task — the task retry logic handles it. After 3 retries, mark session as `FAILED`.

Latency target: AI analysis should complete within 30–60 seconds. Design prompts to be concise. If resume is very long (>3 pages), truncate to first 3 pages for analysis.

2.6 General Notes

AI Match with null job data: When `description` and `skills` are both null, the AI model has minimal context and can only work from `title`, `job_level`, `job_function`, and company name — which is thin. Chosen approach: run AI Match regardless, but pass only available fields to the model and instruct it explicitly to acknowledge the data limitation in the report. Apply a confidence penalty to the displayed score and show a warning badge: *"Limited job data — score based on job title only. For a more accurate match, view the full job post."* If testing after the first AI integration round reveals consistently unreliable scores on null-data jobs, switch to blocking AI Match entirely when both `description` and `skills` are null, replacing the button with: *"Not enough job data to run AI Match. View the original post for full details."* This decision should be revisited after the first round of AI testing — do not finalize the approach before then.

Null field UI behavior (job detail page): Refer to CAREERLY-002 General Notes for the full null field display rules. Summary: `description` and `skills` get styled placeholders with a link to `job_url`; salary fields show *"Salary not disclosed"* if all salary fields are null; `job_level` and `job_function` are omitted entirely if null; minor fields (`vacancy_count`, `emails`, `listing_type`) are silently hidden.

CAREERLY-004 — Notifications Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Notifications — System & New Jobs
Flow ID:       CAREERLY-004
Trigger:       System event (AI complete, resume uploaded, new jobs scraped, account changes)
Entry Point:   Any point in the app — notifications are pushed to the user
Exit Point:    User reads or dismisses the notification; optionally navigates to relevant content
Related Flows: CAREERLY-001 (Auth), CAREERLY-002 (Home Page), CAREERLY-003 (Jobs)
```

1.2 Description

Careerly has two categories of notifications: system notifications (triggered by specific user-related events like AI analysis completion or account changes) and new jobs notifications (triggered at the end of each Celery scrape cycle when relevant new jobs are found for a user). System notifications are dispatched as Django Tasks — they are lightweight, event-driven, and tied directly to user actions. New jobs notifications are dispatched from within the Celery scrape pipeline since they depend on the scrape cycle completing. All notifications are stored in the database for in-app display and marked as read/unread. Push notifications to mobile are delivered via a third-party push service (e.g. Firebase Cloud Messaging).

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Job Seeker	Human	Receives, reads, and dismisses notifications
System	Automated	Creates notification records, marks read/unread
Django Task	Automated	Dispatches lightweight system notifications
Celery Worker	Automated	Dispatches new jobs notifications post-scrape
Push Service (FCM)	Third-Party	Delivers push notifications to mobile devices

1.4 Step-by-Step Bullet Points

System Notifications — Triggers

System notifications are created by a Django Task dispatched via `transaction.on_commit()` immediately after the triggering event. Each trigger maps to a notification type:

Trigger Event	Notification Type	Message
AI Match analysis complete	`AI_MATCH_COMPLETE`	"Your AI Match for [Job Title] at [Company] is ready"
ATS Check analysis complete	`ATS_CHECK_COMPLETE`	"Your resume ATS check is ready"
Resume uploaded successfully	`RESUME_UPLOADED`	"Your resume '[filename]' was uploaded successfully"
Resume parsing failed	`RESUME_PARSE_FAILED`	"We couldn't process your resume. Please try re-uploading."
Account email changed	`ACCOUNT_UPDATED`	"Your email address was updated successfully"
Password changed	`ACCOUNT_UPDATED`	"Your password was changed successfully"
Account suspended by admin	`ACCOUNT_SUSPENDED`	"Your account has been suspended. Contact support."

Flow — System Notification Dispatch

- System event occurs (AI analysis completes, resume uploaded, etc.)
- Celery Worker or View — calls `transaction.on_commit()` to dispatch a Django Task
- Django Task — creates a `Notification` record in the DB:
  - `user` , `type` , `title` , `body` , `related_object_id` (e.g. session ID or job ID), `is_read=False`
- Django Task — checks if the user has a registered FCM push token
  ↳ if push token exists: calls FCM API to send push notification
  ↳ if no push token: skips push, in-app notification is still stored
- Job Seeker — receives push notification on mobile (if token registered)
- Job Seeker — sees unread notification badge on the notification bell in the app
- Job Seeker — opens the notifications panel and sees the notification
- Job Seeker — taps the notification, navigated to the relevant screen (e.g. AI result screen, resume screen)
- System — marks notification as `is_read=True`

Flow — New Jobs Notification Dispatch

- Celery Worker — finishes scrape cycle, collects all newly inserted job IDs
- Celery Worker — queries all active users
- Celery Worker — for each user, checks if any new jobs match their preferences (job titles, locations) or resume keywords
  ↳ if no matches: skips this user — no notification sent
  ↳ if matches found: counts matched jobs, dispatches Django Task for this user
- Django Task — creates one `Notification` record per user:
  - type = `NEW_JOBS` , title = "New jobs for you", body = "X new jobs matching your profile were just added"
- Django Task — sends FCM push notification if token available
- Job Seeker — receives notification: "X new jobs matching your profile were just added"
- Job Seeker — taps notification, navigated to Home page (job listing section, optionally filtered)

Flow — Read / Dismiss Notifications

- Job Seeker — opens notifications panel
- System — returns paginated list of notifications (unread first, then read, sorted by `created_at` desc)
- Job Seeker — taps a notification
- System — marks notification as `is_read=True` , navigates to linked content
- Job Seeker — can also tap "Mark all as read" to bulk-mark all unread notifications
- System — marks all unread notifications for this user as read in a single bulk update

1.5 Validations

Input Validations

Field	Rule	Error Message
FCM token	Max 512 chars, non-empty string	Silently reject malformed tokens — do not error to user
Page number	Positive integer	Invalid requests return empty list

Business Rule Validations

Rule	Condition	Behavior
No duplicate new-jobs notification	New jobs notification already sent this cycle for this user	Skip — one notification per scrape cycle per user
User has no matching jobs	No new jobs match user preferences/resume	Do not create a notification for this user
Notification ownership	User requests another user's notifications	403 Forbidden
Suspended user	User account is suspended	Still receives account-related notifications only

Security Validations

Check	Details
Authentication	JWT required for all notification endpoints
Role-based access	Users only see their own notifications
FCM token registration	Only authenticated user can register their own push token

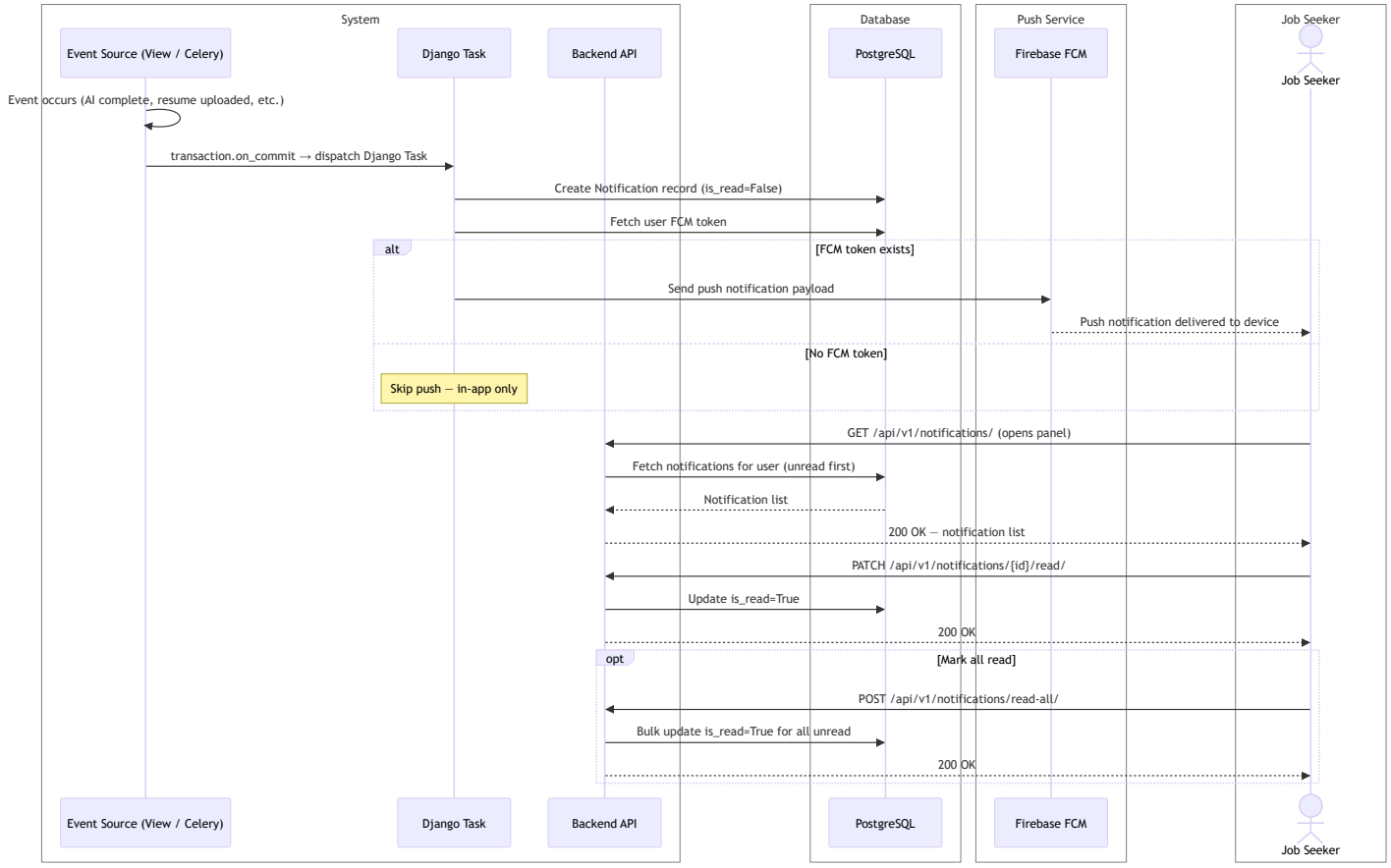
Error Handling

Scenario	System Response
FCM push delivery fails	Log the failure, in-app notification is unaffected — do not retry push more than once
DB write fails for notification	Log the error — do not crash the parent flow (notification failure must never block AI completion or scrape cycle)
User has no active session/token	Skip push silently — in-app only
Notification panel empty	Show "No notifications yet" empty state

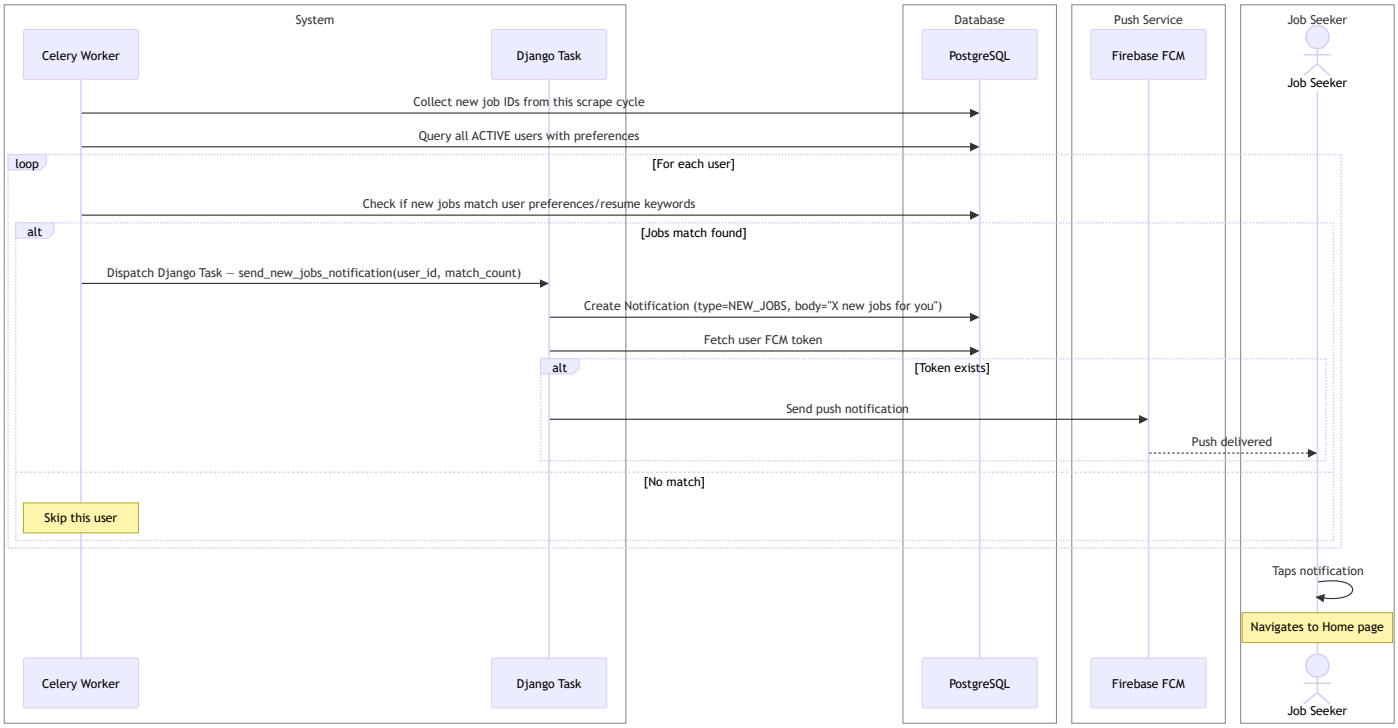
PART 2 — TECHNICAL

2.1 Diagrams

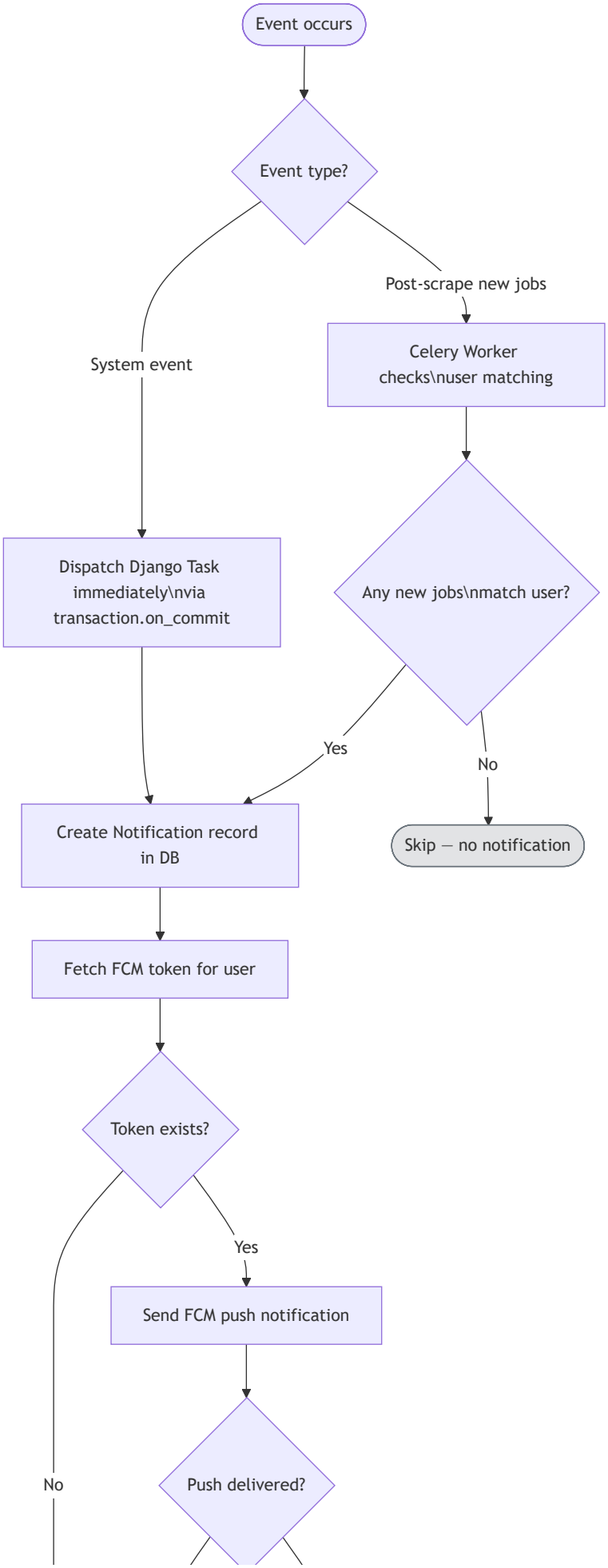
Sequence Diagram — System Notification

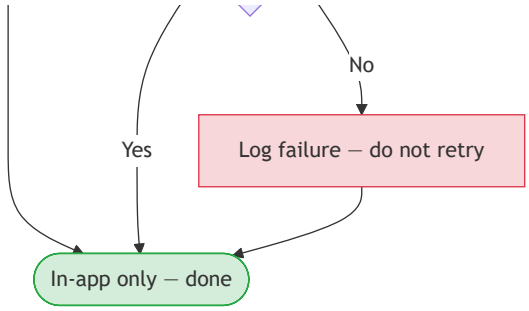


Sequence Diagram — New Jobs Notification (Post-Scrape)



Flowchart — Notification Dispatch Decision


Event occurs
Event type?
System event
Post-scrape new jobs
Celery Worker checks\nuser matching
Dispatch Django Task immediately\nvia transaction.on_commit
Any new jobs\nmatch user?
Yes
No
Skip — no notification
Create Notification record in DB
Fetch FCM token for user
Token exists?
Yes
No
Send FCM push notification
Push delivered?



2.2 Data Models

Model: `Notification`

Purpose: Stores all in-app notifications for a user
Django app: `notifications`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Recipient
`type`	`CharField(choices=NOTIFICATION_TYPE, max_length=30)`	Yes	—	Enum: AI_MATCH_COMPLETE, ATS_CHECK_COMPLETE, RESUME_UPLOADED, RESUME_PARSE_FAILED, NEW_JOBS, ACCOUNT_UPDATED, ACCOUNT_SUSPENDED
`title`	`CharField(max_length=255)`	Yes	—	Short title shown in the panel
`body`	`TextField`	Yes	—	Full notification message
`related_object_id`	`UUIDField(null=True, blank=True)`	No	`null`	ID of related entity (session_id, job_id) for deep link navigation
`related_object_type`	`CharField(max_length=50, null=True, blank=True)`	No	`null`	Type hint for routing: 'ai_session', 'job'
`is_read`	`BooleanField`	No	`False`	Indexed — used for unread badge count
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	Indexed — used for sorting

Model: `PushToken`

Purpose: Stores FCM push tokens per user device
Django app: `notifications`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Token owner
`token`	`CharField(max_length=512, unique=True)`	Yes	—	FCM registration token
`device_type`	`CharField(choices=DEVICE_TYPE, max_length=10)`	Yes	—	Enum: ANDROID, IOS
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—
`last_used_at`	`DateTimeField(auto_now=True)`	Auto	`now`	Updated on each notification attempt

2.3 Table Relationships & Logic

`Notification` is owned by `User`. When the user is deleted, all notifications cascade-delete. Notifications are never deleted by the user — only marked as read. A cleanup Celery Beat task runs weekly to delete notifications older than 90 days.

`PushToken` allows multiple tokens per user (one per device). When a new token is registered, check if it already exists — if yes, update `last_used_at`. If a push delivery fails with an "invalid token" error from FCM, delete that token from the DB automatically.

Unread badge count — computed as:

```
Notification.objects.filter(user=user, is_read=False).count()
```

Cache this count in Redis with key `notifications:unread:{user_id}`, TTL 5 minutes. Invalidate on new notification creation and on mark-read operations.

`related_object_id` + `related_object_type` — used for deep linking. When the user taps a notification, the frontend reads these fields and routes to the correct screen:

- `ai_session` → navigate to AI session result screen
- `job` → navigate to job detail screen
- `null` → no navigation, just mark as read

New jobs notification deduplication — before creating a `NEW_JOBS` notification, check if one was already created for this user in the last 15 minutes (one scrape cycle). Use:

```
Notification.objects.filter(
    user=user,
    type='NEW_JOBS',
    created_at__gte=now() - timedelta(minutes=15)
).exists()
```

If one exists, skip creation.

2.4 API Endpoints

Method	Endpoint	Auth	Role	Request Body / Params	Response	Desc
`GET`	`/api/v1/notifications/`	Yes	Job Seeker	`?page=N`	`200` — `{notifications[], unread_count, pagination}`	List u notific
`PATCH`	`/api/v1/notifications/{id}/read/`	Yes	Job Seeker	—	`200`	Mark notific as rea
`POST`	`/api/v1/notifications/read-all/`	Yes	Job Seeker	—	`200`	Mark read
`POST`	`/api/v1/notifications/push-token/`	Yes	Job Seeker	`{token, device_type}`	`200`	Regis updat FCM token
`DELETE`	`/api/v1/notifications/push-token/`	Yes	Job Seeker	`{token}`	`204`	Remo push on log

2.5 Developer Notes

🔵 Backend Developer (Django)

- Create a `notifications` Django app with `Notification` and `PushToken` models.
- Write a utility function `create_notification(user, type, title, body, related_object_id=None, related_object_type=None)` — used everywhere to create notifications consistently.
- Write a utility function `dispatch_push(user, title, body, data={})` — fetches all push tokens for the user, calls FCM for each. On `invalid token` FCM error, auto-delete that token.
- Use `firebase-admin` Python SDK for FCM: `pip install firebase-admin`. Initialize with service account credentials in Django settings.
- Use `transaction.on_commit()` before every Django Task dispatch in views and Celery tasks.
- New jobs notification dispatch: run inside the Celery scrape task after all platforms are scraped. Use `User.objects.filter(status='ACTIVE').select_related('preferences')` and batch process. Do not run one DB query per user — fetch all users and their preferences, then filter in Python or use a single optimized query.
- Cache unread count per user in Redis. Invalidate on `post_save` signal of `Notification` and after mark-read operations.
- Notification cleanup Celery Beat task: `Notification.objects.filter(created_at__lt=now()-90days).delete()` — runs weekly.
- On user logout: call `DELETE /api/v1/notifications/push-token/` to remove the token — ensure this is called from the logout endpoint automatically.

🟢 Frontend Developer (React)

- Notification bell icon in the top nav bar with an unread count badge (red dot or number).
- Fetch unread count on app load and cache it. Refresh when the notifications panel is opened.
- Notifications panel: slide-in drawer or dropdown — `GET /api/v1/notifications/` on open.
- List items: title, body, relative timestamp ("2 minutes ago"), read/unread visual state (unread = highlighted background or bold text).
- On item click: call `PATCH /api/v1/notifications/{id}/read/` and navigate using `related_object_type` + `related_object_id`.
- "Mark all as read" button at the top of the panel.
- On web, use browser `Notification` API for push if the user grants permission — fallback to in-app only if not granted.
- Register service worker for web push (optional for MVP — prioritize mobile push).

🟡 Mobile Developer (Flutter)

- Use `firebase_messaging` Flutter package for FCM integration.
- On app launch (after login): call `FirebaseMessaging.instance.getToken()` to get the FCM token and `POST /api/v1/notifications/push-token/` to register it.
- On logout: `DELETE /api/v1/notifications/push-token/` with the current token.
- Handle three notification states:
  - **Foreground**: show in-app banner (`flutter_local_notifications`).
  - **Background/terminated**: FCM delivers it as a system notification. On tap, open the app and navigate using `data.related_object_type` + `data.related_object_id`.
- Deep link routing: implement a `NotificationRouter` that reads `related_object_type` and calls the appropriate screen navigator.
- Notification panel: `ListView.builder` with `NotificationTile` widgets. Unread tiles have a different background color.
- Badge count: display on the notification bell icon using `badges` Flutter package.
- Request notification permissions on iOS explicitly (`FirebaseMessaging.instance.requestPermission()`).

🟣 AI Engineer

N/A for this flow. Notifications are triggered by AI session completions, but the notification logic itself is system-level only.

CAREERLY-005 — Profile Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Profile — View & Edit
Flow ID:       CAREERLY-005
Trigger:       User navigates to their profile page or edit profile page
Entry Point:   Profile tab / Profile screen
Exit Point:    Profile viewed or updated successfully
Related Flows: CAREERLY-001 (Auth — resume uploaded at setup), CAREERLY-003 (Jobs — AI sessions feed profile stats)
```

1.2 Description

The profile flow covers both viewing and editing a job seeker's profile. The profile view is a summary dashboard showing the user's personal info, computed stats (average ATS score, average suitability score, total jobs viewed), their uploaded resumes, and a list of jobs they have viewed. The edit profile page allows the user to update their personal information and manage multiple resume files. The biography field supports markdown so users can format their bio with structure. This flow does not cover the AI analysis sessions themselves (CAREERLY-003) but reads their aggregated results.

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Job Seeker	Human	Views their profile, edits personal info, manages resumes
System	Automated	Computes and serves stats, validates inputs, stores updates

1.4 Step-by-Step Bullet Points

Route 1 — View Profile

- Job Seeker — navigates to the profile screen
- System — fetches user data, computed stats, resumes, and viewed jobs list
- System — computes stats:
  - Average ATS score: average of all completed ATS_CHECK sessions
  - Average suitability score: average of all completed JOB_MATCH sessions
  - Total jobs viewed: count of JobView records for this user
- Job Seeker — sees:
  - Profile image (or placeholder if not set)
  - Full name
  - Current position/title
  - Average ATS score (with verdict color)
  - Average suitability score (with verdict color)
  - Total jobs viewed count
  - Biography (rendered as markdown)
  - List of uploaded resumes (filename, upload date, parse status)
  - Paginated list of viewed jobs (title, company, date viewed)

Route 2 — Edit Profile

- Job Seeker — taps "Edit Profile"
- System — loads the current profile data into the edit form

- Job Seeker — updates any combination of:
  - Profile image (upload new image)
  - Full name
  - Email address
  - Phone number
  - Position/title
  - Country
  - Biography (markdown input)
- Job Seeker — taps Save
- System — validates all updated fields
  ↳ if validation fails: highlights invalid fields, does not save
- System — if email was changed:
  - Updates email
  - Dispatches Django Task to send "Your email has been updated" system notification
  - Does NOT require re-verification (email is already trusted from registration)
- System — saves updated profile
- Job Seeker — sees updated profile

Route 3 — Manage Resumes

- Job Seeker — navigates to the resumes section (within edit profile or standalone)
- System — lists all uploaded resumes for this user (filename, size, upload date, parse status, default badge)
- Job Seeker — can upload a new resume (PDF or DOCX, max 5MB)
- System — validates the file
  ↳ if invalid type or size: shows error, does not upload
- System — saves the resume, dispatches Celery Task to parse it
- System — if this is the user's first resume, sets it as default automatically
- Job Seeker — can set any resume as the default (used for AI features)
- System — updates `is_default` flag (sets new default, removes old default)
- Job Seeker — can delete a resume
  ↳ if deleting the default resume and other resumes exist: system auto-sets the most recently uploaded as the new default
  ↳ if deleting the only resume: warns user "This is your only resume. Deleting it will disable AI features."
  ↳ if user confirms: deletes it — AI features will be unavailable until a new resume is uploaded

1.5 Validations

Input Validations

Field	Rule	Error Message
Full name	Required, 2–100 chars, letters and spaces only	"Please enter a valid full name"
Email	Valid email format, required	"Please enter a valid email address"
Phone number	Optional, valid international format (E.164)	"Please enter a valid phone number"
Position	Optional, max 100 chars	"Position must be under 100 characters"
Country	Optional, must be from a predefined list	"Please select a valid country"
Biography	Optional, max 2000 chars, markdown allowed	"Biography must be under 2000 characters"
Profile image	Optional, JPG/PNG only, max 2MB	"Only JPG and PNG images are accepted" / "Image must be under 2MB"
Resume file	PDF or DOCX only, max 5MB	"Only PDF and DOCX files are accepted" / "File size must be under 5MB"

Business Rule Validations

Rule	Condition	Behavior
Resume limit	More than 10 resumes uploaded	Block upload — "You can have a maximum of 10 resumes"

Rule	Condition	Behavior
Default resume	Deleting the default resume	Auto-assign next most recent as default, or warn if it's the last one
Email uniqueness	New email already used by another account	Block — "This email is already in use"

Security Validations

Check	Details
Authentication	JWT required — profile is private
Role-based access	Users can only edit their own profile
File MIME type	Validate server-side — reject spoofed extensions
Profile image	Strip EXIF data before storing

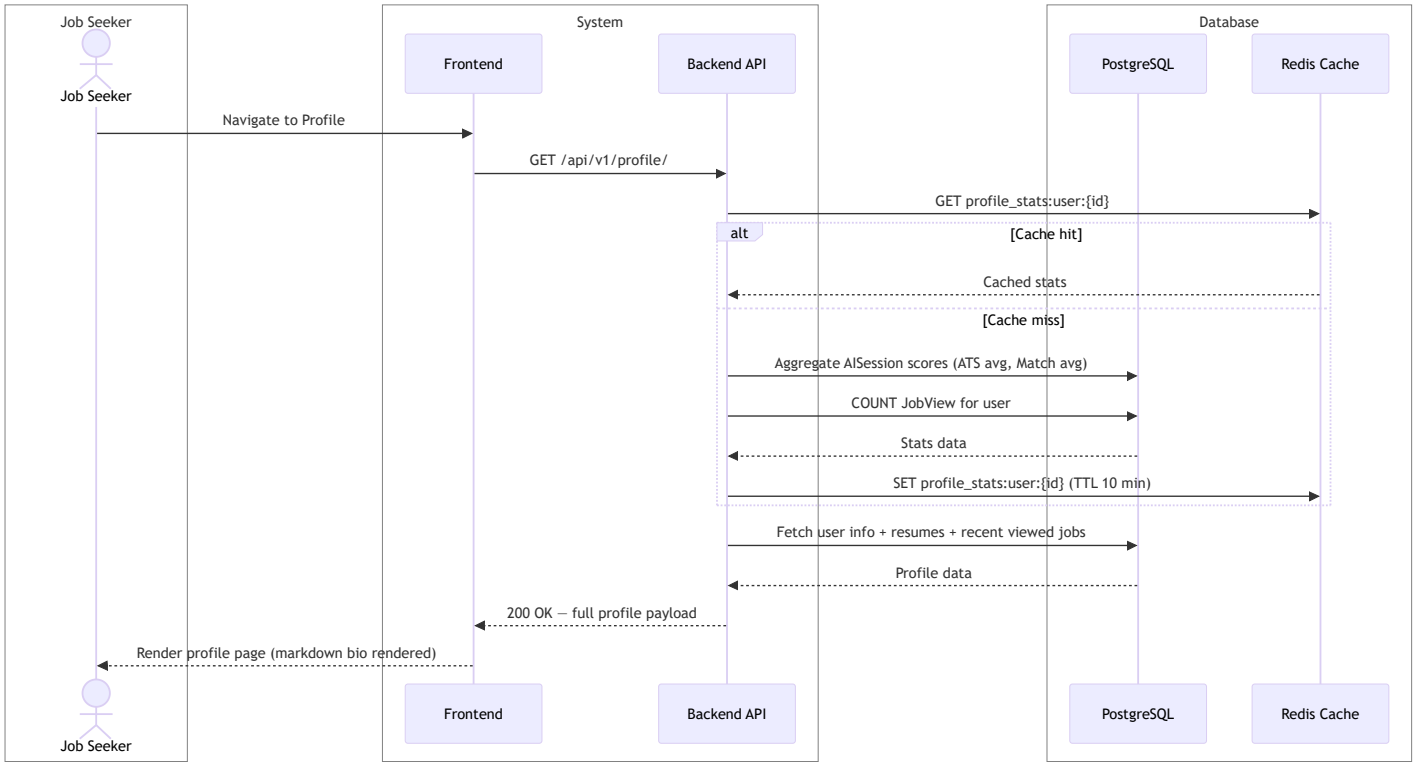
Error Handling

Scenario	System Response
Image upload fails	Show "Image upload failed. Try again." — preserve other form changes
Resume upload fails	Show "Resume upload failed. Try again."
Server error on save	Show generic error — preserve form data
Stats computation returns null	Show "—" or "N/A" for stats with no data yet
No viewed jobs yet	Show "You haven't viewed any jobs yet" empty state

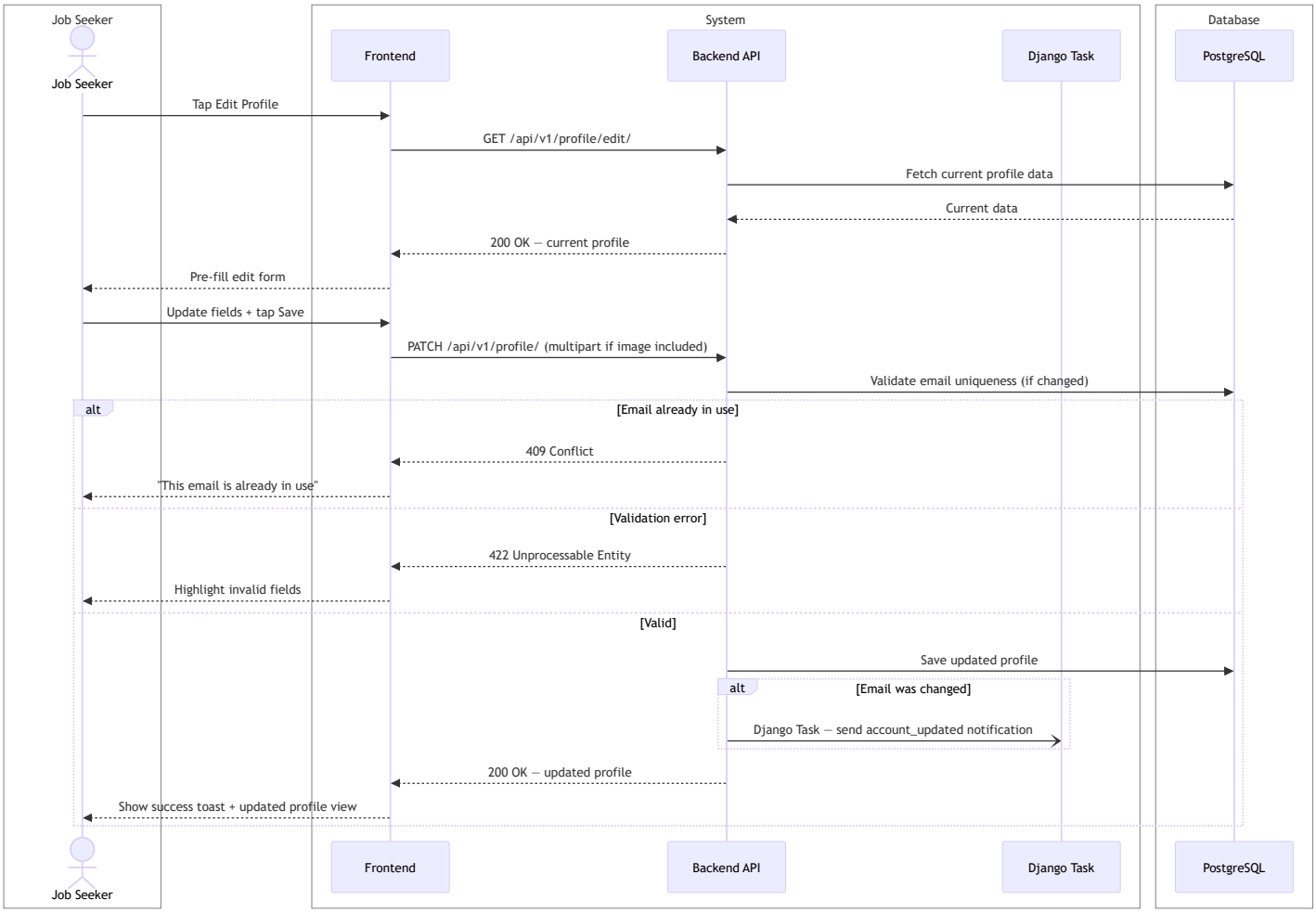
PART 2 — TECHNICAL

2.1 Diagrams

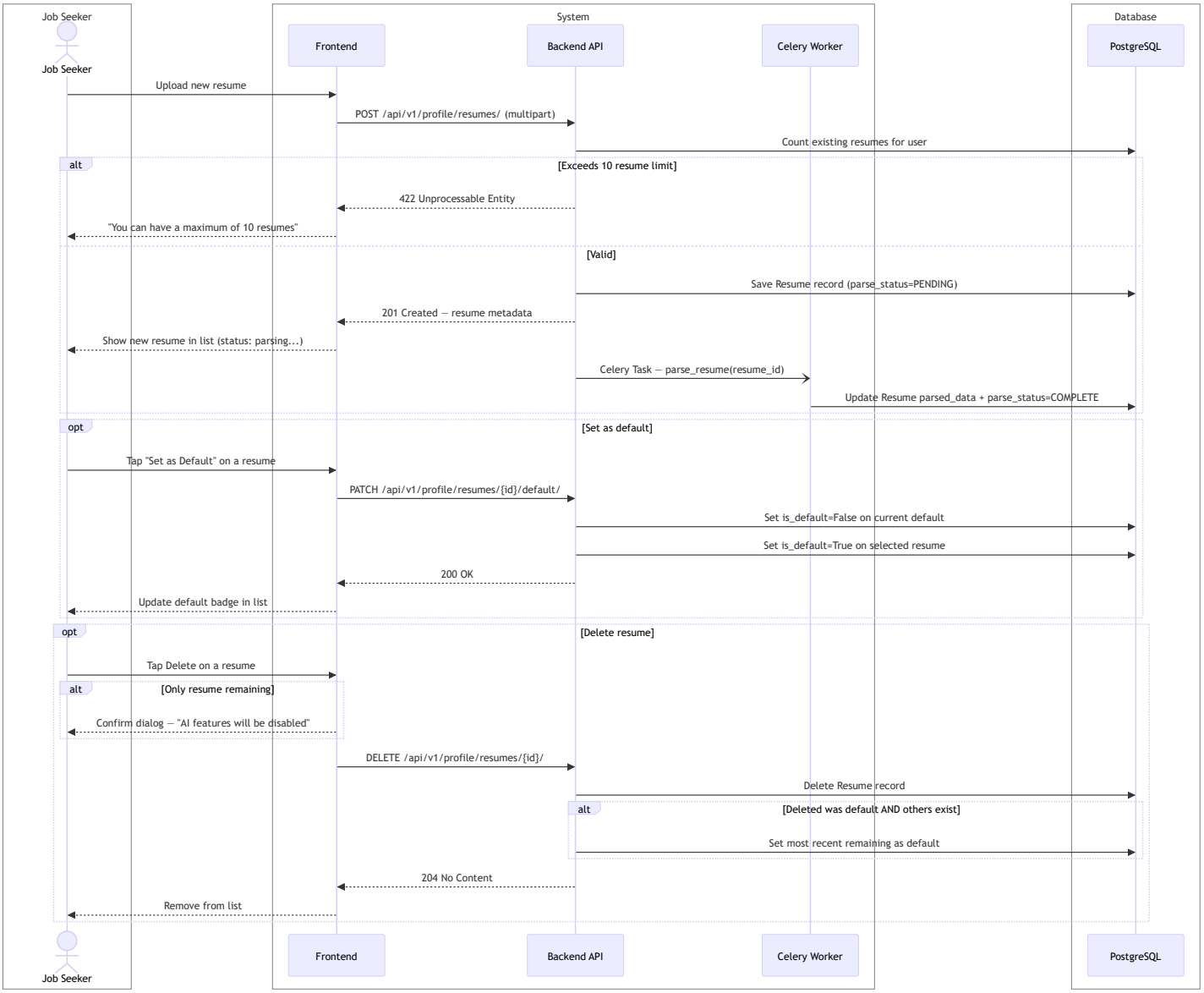
Sequence Diagram — View Profile



Sequence Diagram — Edit Profile



Sequence Diagram — Manage Resumes



2.2 Data Models

Model: `UserProfile`

Purpose: Extended profile data for a job seeker — separate from the auth `User` model

Django app: `accounts`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`OneToOneField(User, on_delete=CASCADE)`	Yes	—	One profile per user
`full_name`	`CharField(max_length=100)`	Yes	—	Display name
`phone_number`	`CharField(max_length=20, null=True, blank=True)`	No	`null`	E.164 format
`position`	`CharField(max_length=100, null=True, blank=True)`	No	`null`	Current or desired job title
`country`	`CharField(max_length=100, null=True, blank=True)`	No	`null`	From a validated country list
`biography`	`TextField(max_length=2000, null=True, blank=True)`	No	`null`	Supports markdown

Field	Django Field Type	Required	Default	Notes
`profile_image`	`ImageField(upload_to='profiles/', null=True, blank=True)`	No	`null`	Stored in S3 / media
`updated_at`	`DateTimeField(auto_now=True)`	Auto	`now`	—

Model: `Resume` (referenced from CAREERLY-003)

Purpose: Resume files uploaded by a user
Django app: `accounts`

(Full definition in CAREERLY-003 — reproduced here for completeness)

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	Owner
`file`	`FileField(upload_to='resumes/')`	Yes	—	S3 or local
`original_filename`	`CharField(max_length=255)`	Yes	—	Shown in UI
`file_size`	`PositiveIntegerField`	Yes	—	Bytes
`parsed_data`	`JSONField(null=True, blank=True)`	No	`null`	Extracted: skills, titles, experience
`parse_status`	`CharField(choices=PARSE_STATUS, max_length=20)`	Yes	`PENDING`	PENDING, COMPLETE, FAILED
`is_default`	`BooleanField`	No	`False`	One default per user
`uploaded_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

2.3 Table Relationships & Logic

`UserProfile` has a `OneToOneField` to `User`. It is created automatically via a `post_save` signal on `User` — every user gets a `UserProfile` at creation time with empty optional fields.

`Resume` has a `ForeignKey` to `User`. Multiple resumes per user are allowed (max 10 enforced at the API level). The `is_default` flag has a `UniqueConstraint` filtered on `is_default=True` to enforce one default per user at the DB level.

Stats computation:

- Average ATS score: `AISession.objects.filter(user=user, session_type='ATS_CHECK', status='COMPLETE').aggregate(Avg('score'))['score__avg']`
- Average suitability score: `AISession.objects.filter(user=user, session_type='JOB_MATCH', status='COMPLETE').aggregate(Avg('score'))['score__avg']`
- Total jobs viewed: `JobView.objects.filter(user=user).count()`

All three are cached in Redis under `profile_stats:user:{id}` as a single JSON object, TTL 10 minutes. Invalidated when:

- A new `AISession` reaches `COMPLETE` status (via `post_save` signal)
- A new `JobView` is created (via `post_save` signal)

Viewed jobs list on profile — paginated, sorted by `viewed_at` desc. Uses `JobView.objects.filter(user=user).select_related('job').order_by('-viewed_at')`.

Biography markdown — stored as raw markdown text in the DB. Never pre-rendered on the backend. Rendered exclusively on the frontend. Max 2000 chars validated at the API level.

Profile image — strip EXIF metadata server-side using `Pillow` before saving. Resize to max 500×500px to prevent storing unnecessarily large images. Store processed version only.

Resume default reassignment — implemented as a `post_delete` signal on `Resume`:

```python
@receiver(post_delete, sender=Resume)
def reassign_default_resume(sender, instance, **kwargs):
    if instance.is_default:
        next_resume = Resume.objects.filter(
            user=instance.user
        ).order_by('-uploaded_at').first()
        if next_resume:
            next_resume.is_default = True
            next_resume.save()
```

2.4 API Endpoints

Method	Endpoint	Auth	Role	Request Body / Params
`GET`	`/api/v1/profile/`	Yes	Job Seeker	—
`PATCH`	`/api/v1/profile/`	Yes	Job Seeker	`multipart: {full_name, email, phone, position, country, biography, profil`
`GET`	`/api/v1/profile/resumes/`	Yes	Job Seeker	—
`POST`	`/api/v1/profile/resumes/`	Yes	Job Seeker	`multipart: {resume_file}`
`PATCH`	`/api/v1/profile/resumes/{id}/default/`	Yes	Job Seeker	—
`DELETE`	`/api/v1/profile/resumes/{id}/`	Yes	Job Seeker	—
`GET`	`/api/v1/profile/viewed-jobs/`	Yes	Job Seeker	`?page=N`

2.5 Developer Notes

🔵 Backend Developer (Django)

- Create `UserProfile` via `post_save` signal on `User` model — use `get_or_create` to be safe.
- `PATCH /api/v1/profile/` : use a partial serializer (`partial=True` in DRF). Only update fields that are present in the request — do not overwrite unspecified fields with null.
- Profile image processing: use `Pillow` to strip EXIF and resize to 500×500 max before saving. Do this in the serializer's `validate_profile_image` method.
- Resume upload: `FileField` with `upload_to='resumes/{user_id}/'` — organize by user ID. Validate MIME type using `python-magic` (not just extension).
- `parse_resume` Celery task: extract text from PDF using `pdfplumber` or `PyMuPDF` . Extract from DOCX using `python-docx` . Send extracted text to AI service for parsing. Store result in `Resume.parsed_data` as JSON.
- Cache invalidation: add `post_save` signal on `AISession` (when `status` changes to `COMPLETE`) to `cache.delete(f'profile_stats:user:{instance.user_id}')` .
- Stats aggregation: use a single DB call that returns all three stats together — avoid three separate queries.
- Viewed jobs endpoint: `JobView.objects.filter(user=user).select_related('job').order_by('-viewed_at')` . Use `only('job__id', 'job__title', 'job__company', 'viewed_at')` to avoid overfetching.

🟢 Frontend Developer (React)

- Profile page sections: top card (image, name, position, stats row), bio section (render with `react-markdown`), resumes section (file cards), viewed jobs section (paginated list).
- Stats row: three stat blocks — "Avg ATS Score" (with color-coded score), "Avg Match Score", "Jobs Viewed". Show "—" if no data.
- Edit form: use controlled inputs pre-filled from `GET /api/v1/profile/edit/`. Only send changed fields in the `PATCH` request.
- Biography input: use a markdown-aware textarea (e.g. `react-mde` or `@uiw/react-md-editor`) with a live preview toggle.
- Resume list: each item shows filename, upload date, parse status badge (Parsing... / Ready / Failed), "Set as Default" button (hidden on current default), and "Delete" with confirmation dialog.
- Profile image upload: click-to-upload circle avatar. Show preview before saving.
- Viewed jobs list: `JobCard` components (reuse from Home/Jobs). Paginate with "Load More" button.

🟡 Mobile Developer (Flutter)

- Profile screen: `CustomScrollView` with `SliverAppBar` showing the profile image, then `SliverList` for sections.
- Profile image: `CircleAvatar` with `CachedNetworkImage`. On tap, open image picker using `image_picker` package.
- Biography: use `flutter_markdown` for rendering. In edit mode, use a plain `TextField` with a preview toggle.
- Stats row: `Row` of three `StatCard` widgets with color-coded score values.
- Resume list: `ListView` with swipe-to-delete (using `Dismissible` widget) and a "Set as Default" option in a long-press context menu.
- Edit form: same fields as web. Use `TextInputType.phone` for phone number, `TextInputType.emailAddress` for email.
- Navigation: profile tab is one of the bottom nav tabs — use `IndexedStack` to preserve scroll state.

🟣 AI Engineer

- Resume parsing Celery task (`parse_resume`):
  - Input: `resume_id` — fetch file from storage, extract text.
  - PDF text extraction: use `pdfplumber` — handles multi-column layouts better than PyPDF2.
  - DOCX text extraction: use `python-docx`.
  - Send extracted text to AI parsing model. Expected output JSON: `{skills: [], job_titles: [], years_experience: int, education_level: str}`.
  - Store in `Resume.parsed_data`. Set `parse_status=COMPLETE`.
  - On failure: set `parse_status=FAILED`, dispatch Django Task to send `RESUME_PARSE_FAILED` notification.
- This parsed data feeds: home page recommendations (CAREERLY-002), AI Match (CAREERLY-003), and ATS Check (CAREERLY-003).
- Keep the parsing model fast — target under 10 seconds. If resume text is very long, truncate to first 5000 tokens before sending.

CAREERLY-006 — Subscription Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Subscription — Plans & Feature Gating
Flow ID:       CAREERLY-006
Trigger:       User attempts to use a gated feature, or navigates to the subscription/upgrade screen
Entry Point:   Any gated feature CTA, or the Subscription screen in settings
Exit Point:    User is subscribed to a plan and has access to its features
Related Flows: CAREERLY-003 (Jobs — AI features are gated), CAREERLY-001 (Auth — plan assigned at registration)
```

1.2 Description

Careerly uses a subscription model to govern access to premium features. Every user is assigned a plan — new users start on the FREE plan by default. Plans define a set of feature flags and usage limits (e.g. number of AI Match sessions per month). The feature configuration is intentionally flexible — limits and flags are stored in the database and can be changed by an admin without a code deployment. This flow covers plan assignment, upgrade, and feature

gate enforcement. Payment processing is out of scope for this version — the subscription record is created manually or through a payment integration to be defined later.

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Job Seeker	Human	Views plans, upgrades subscription, uses features within limits
Admin	Human	Configures plans and feature limits via admin dashboard
System	Automated	Enforces feature gates, tracks usage, assigns default plan on registration

1.4 Step-by-Step Bullet Points

Route 1 — Plan Assignment at Registration

- System — when a new user completes registration and setup, automatically assigns the FREE plan
- System — creates a `UserSubscription` record with `plan=FREE`, `status=ACTIVE`, `start_date=today`

Route 2 — View Plans & Upgrade

- Job Seeker — navigates to the Subscription screen
- System — fetches all available plans with their features and limits
- Job Seeker — sees the current plan (highlighted) and available upgrade options
- Job Seeker — selects a plan to upgrade to
- System — (payment processing TBD) — for now, records the plan change directly
- System — updates `UserSubscription` with new plan and `start_date=today`
- Job Seeker — immediately has access to the new plan's features

Route 3 — Feature Gate Enforcement

- Job Seeker — attempts to use a feature (e.g. AI Match)
- System — checks the user's active plan
- System — checks if the feature is enabled for this plan
  ↳ if feature not in plan: blocks action, shows upgrade prompt
- System — checks if the user has remaining usage for this feature (if it has a limit)
  ↳ if usage limit reached: blocks action, shows "You've reached your limit for this month. Upgrade to continue."
  ↳ if within limit: allows action, increments usage counter

1.5 Validations

Business Rule Validations

Rule	Condition	Behavior
Default plan	New user registered	Auto-assign FREE plan
Feature access	Feature not included in user's plan	Block + show upgrade prompt
Usage limit	User has hit monthly limit for a feature	Block + show upgrade prompt with limit details
Downgrade	User downgrades to a lower plan	Usage resets at next cycle, existing sessions preserved

Security Validations

Check	Details
Authentication	JWT required

Check	Details
Feature gate checks	Always enforced server-side — never trust client-side gating alone
Plan configuration	Only admins can create or modify plans

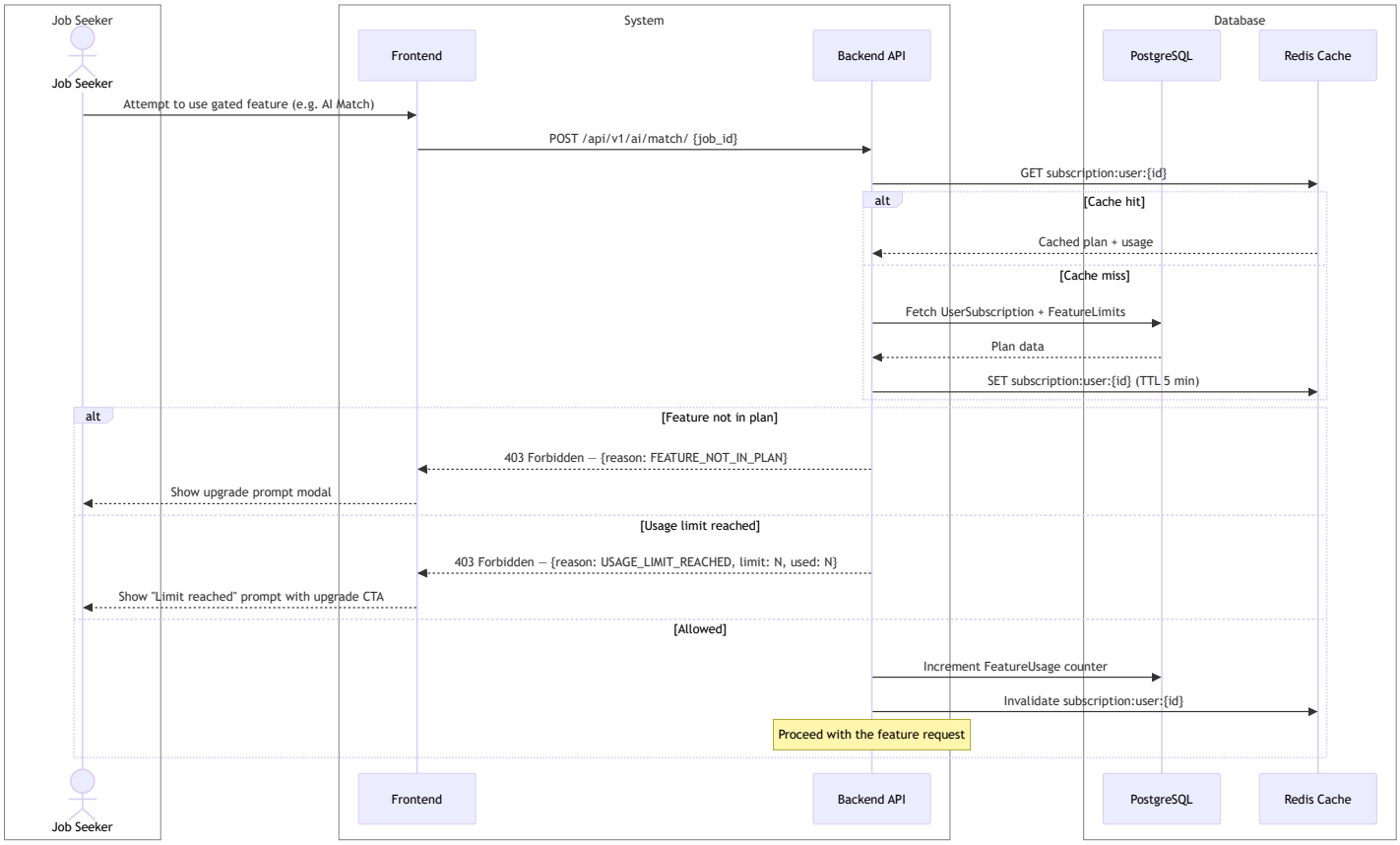
Error Handling

Scenario	System Response
Subscription record missing	Auto-create with FREE plan — do not error
Plan config not found	Fall back to most restrictive defaults, alert admin

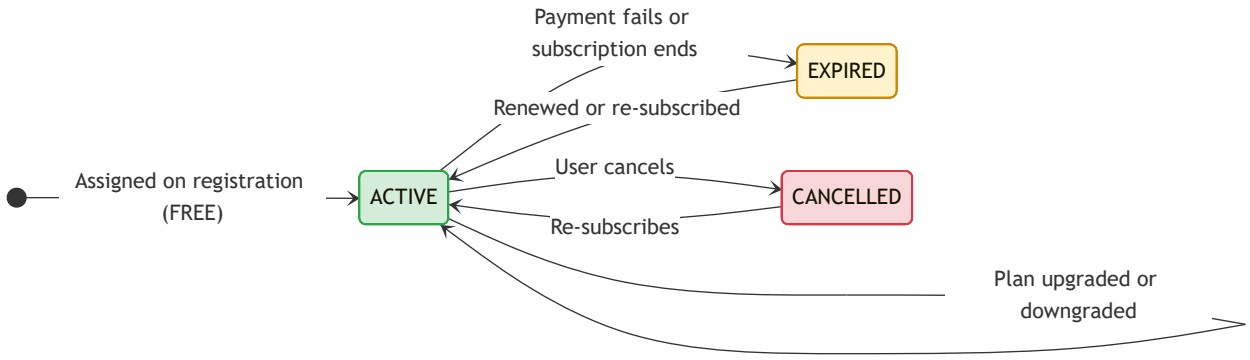
PART 2 — TECHNICAL

2.1 Diagrams

Sequence Diagram — Feature Gate Check



State Diagram — Subscription Status



2.2 Data Models

Model: `Plan`

Purpose: Defines a subscription tier and its feature configuration
Django app: `subscriptions`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`name`	`CharField(max_length=50, unique=True)`	Yes	—	e.g. FREE, PRO, PREMIUM
`display_name`	`CharField(max_length=100)`	Yes	—	Shown in UI
`price_monthly`	`DecimalField(max_digits=8, decimal_places=2)`	Yes	`0.00`	0 for free plan
`is_active`	`BooleanField`	No	`True`	Inactive plans are hidden from upgrade screen
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Model: `PlanFeature`

Purpose: Configurable feature flags and limits per plan — admin-editable
Django app: `subscriptions`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`plan`	`ForeignKey(Plan, on_delete=CASCADE)`	Yes	—	Which plan this feature belongs to
`feature_key`	`CharField(max_length=100)`	Yes	—	e.g. `ai_match`, `ats_check`, `saved_jobs`. Indexed.
`is_enabled`	`BooleanField`	Yes	`True`	Whether the feature is available at all
`monthly_limit`	`PositiveIntegerField(null=True, blank=True)`	No	`null`	Null = unlimited. e.g. 5 for AI Match on FREE

Unique constraint: `unique_together = [('plan', 'feature_key')]`

Model: `UserSubscription`

Purpose: The active subscription record for a user
Django app: `subscriptions`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`OneToOneField(User, on_delete=CASCADE)`	Yes	—	One active subscription per user

Field	Django Field Type	Required	Default	Notes
`plan`	`ForeignKey(Plan, on_delete=PROTECT)`	Yes	—	PROTECT — do not delete plans with active subscribers
`status`	`CharField(choices=SUB_STATUS, max_length=20)`	Yes	`ACTIVE`	Enum: ACTIVE, EXPIRED, CANCELLED
`start_date`	`DateField`	Yes	`today`	When this subscription period started
`end_date`	`DateField(null=True, blank=True)`	No	`null`	Null for ongoing; set for fixed-term
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—
`updated_at`	`DateTimeField(auto_now=True)`	Auto	`now`	—

Model: `FeatureUsage`

Purpose: Tracks monthly usage of limited features per user
Django app: `subscriptions`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`user`	`ForeignKey(User, on_delete=CASCADE)`	Yes	—	The user
`feature_key`	`CharField(max_length=100)`	Yes	—	Same key as PlanFeature.feature_key
`month`	`DateField`	Yes	—	First day of the month — e.g. 2025-06-01. Indexed.
`count`	`PositiveIntegerField`	No	`0`	Incremented on each use

Unique constraint: `unique_together = [('user', 'feature_key', 'month')]`

2.3 Table Relationships & Logic

`Plan` → `PlanFeature` is one-to-many. Each plan has multiple feature entries. Admin configures these via Django admin.

`UserSubscription` is a `OneToOneField` to `User`. Created automatically when the user completes setup. If the record is missing (e.g. legacy users), auto-create with FREE plan on access.

`FeatureUsage` tracks per-user, per-feature, per-month usage. The `month` field is always the first day of the current month (e.g. `date.today().replace(day=1)`). Use `get_or_create` when incrementing:

```
usage, _ = FeatureUsage.objects.get_or_create(
    user=user,
    feature_key=feature_key,
    month=date.today().replace(day=1),
    defaults={'count': 0}
)
usage.count = F('count') + 1
usage.save(update_fields=['count'])
```

Using `F('count') + 1` is atomic — prevents race conditions.

Feature gate check logic (implemented as a reusable utility):

```python
def can_use_feature(user, feature_key) -> tuple[bool, str]:
    sub = UserSubscription.objects.select_related('plan').get(user=user)
    feature = PlanFeature.objects.get(plan=sub.plan, feature_key=feature_key)

    if not feature.is_enabled:
        return False, 'FEATURE_NOT_IN_PLAN'

    if feature.monthly_limit is not None:
        month = date.today().replace(day=1)
        usage = FeatureUsage.objects.filter(
            user=user, feature_key=feature_key, month=month
        ).first()
        current_count = usage.count if usage else 0
        if current_count >= feature.monthly_limit:
            return False, 'USAGE_LIMIT_REACHED'

    return True, 'ALLOWED'
```

Cache the plan + feature config in Redis per user. Invalidate when subscription or plan config changes.

Feature keys (initial set — admin can add more):

Key	Description
`ai_match`	AI Match feature
`ats_check`	Check Your Resume / ATS
`saved_jobs`	Save jobs for later
`job_recommendations`	Personalized recommendations

2.4 API Endpoints

Method	Endpoint	Auth	Role	Request Body / Params	Response	Description
`GET`	`/api/v1/subscriptions/plans/`	No	—	—	`200` — list of active plans with features	Public plans listing
`GET`	`/api/v1/subscriptions/me/`	Yes	Job Seeker	—	`200` — current subscription + usage	Get own subscription status
`POST`	`/api/v1/subscriptions/upgrade/`	Yes	Job Seeker	`{plan_id}`	`200` — updated subscription	Upgrade plan (payment TBD)

2.5 Developer Notes

🔵 Backend Developer (Django)

- Create a reusable `can_use_feature(user, feature_key)` utility in `subscriptions/utils.py`. Use this as a check at the start of any gated API view.
- Create a DRF permission class `HasFeatureAccess(feature_key)` that calls `can_use_feature` — use it as a permission on gated views: `permission_classes = [IsAuthenticated, HasFeatureAccess('ai_match')]`.
- `post_save` signal on `User` (status changes to `ACTIVE`): create `UserSubscription` with FREE plan.
- Use `select_related('plan')` when fetching `UserSubscription` to avoid N+1 on plan lookups.
- Cache `subscription:user:{id}` in Redis as a dict of `{plan_name, features: {key: {is_enabled, monthly_limit, used}}}`. TTL 5 minutes. Invalidate on subscription change.
- Monthly usage reset: Celery Beat task on the 1st of each month — not needed since usage is tracked by `month` field. No reset job required — the `get_or_create` on a new month key naturally starts at 0.
- Admin: register `Plan`, `PlanFeature`, `UserSubscription`, `FeatureUsage` in Django admin. Use `inline` for `PlanFeature` under `Plan` for easy editing.

🟢 Frontend Developer (React)

- Subscription screen: plan cards side-by-side. Current plan has "Current Plan" badge. Other plans have "Upgrade" CTA.
- Each plan card: name, price, feature list (checkmarks for included, crossed for excluded), monthly limits shown inline ("5 AI Matches/month").
- When a feature gate is hit (API returns 403 `FEATURE_NOT_IN_PLAN` or `USAGE_LIMIT_REACHED`): show a modal with the reason and an "Upgrade" button that routes to the subscription screen.
- Current usage display: on the subscription page, show a usage bar for limited features ("3 of 5 AI Matches used this month").
- Feature flags: also fetch from `GET /api/v1/subscriptions/me/` on app load and store in global state — use this to conditionally show/hide feature buttons in the UI. This is a UX enhancement only — server-side gating is the enforcement layer.

🟡 Mobile Developer (Flutter)

- Subscription screen: `PageView` or `Column` of plan cards with feature comparison.
- Gate enforcement: intercept 403 responses in the global HTTP interceptor — if `reason` is `FEATURE_NOT_IN_PLAN` or `USAGE_LIMIT_REACHED`, show a `BottomSheet` with upgrade prompt.
- Cache the user's plan and feature access in local state after login — refresh on subscription change.
- Usage bars: `LinearProgressIndicator` for each limited feature.

🟣 AI Engineer

- AI features (`ai_match` , `ats_check`) are gated by this subscription system.
- Before dispatching an AI Celery task, the backend checks `can_use_feature` . If allowed, it increments usage and then dispatches the task.
- If usage limit is reached mid-session (edge case — user hits limit between check and completion), the session is still completed — do not cancel in-flight AI tasks.
- No AI involvement in the subscription logic itself.

2.6 General Notes

- Payment integration details to be added once provider is selected
- Plan names and limits to be finalized with product team

CAREERLY-007 — Monitoring Flow

PART 1 — ANALYSIS

1.1 Flow Title & Metadata

```
Flow Name:     Monitoring — Admin Dashboard & System Analytics
Flow ID:       CAREERLY-007
Trigger:       Admin logs into the admin dashboard and navigates to monitoring
Entry Point:   Django Admin — Monitoring section
Exit Point:    Admin views metrics, takes action if needed
Related Flows: All flows — monitoring reads data from all other flows
```

1.2 Description

The monitoring flow provides the admin with a real-time and historical view of Careerly's health and usage. It extends Django's built-in admin with custom views that render area charts, stat cards, and color-coded health indicators. The dashboard covers platform health (scraper success rates), user activity (registrations, active users, jobs viewed), AI usage (resumes checked, ATS scores, match scores), and subscription metrics (conversion rates). Data is aggregated from the existing models — no separate analytics database is needed at this stage. This flow is admin-only and does not affect any user-facing features.

1.3 Actors / User Roles

Role	Type	Responsibilities in this flow
Admin	Human	Views metrics, monitors system health, identifies anomalies
System	Automated	Aggregates metrics, caches results, serves dashboard data
Celery Beat	Automated	Runs periodic aggregation tasks to pre-compute heavy metrics

1.4 Step-by-Step Bullet Points

Dashboard Load

- Admin — navigates to the monitoring section in Django admin
- System — loads the custom admin monitoring view
- System — checks Redis cache for pre-computed metrics
  ↳ if cache hit: returns cached metrics immediately
  ↳ if cache miss: computes metrics from DB and caches them
- Admin — sees the dashboard with:
  - Platform health cards (scraper status per platform — green/yellow/red)
  - Key stats row (total users, new users today, total jobs, new jobs today)
  - Area chart: jobs viewed throughout the week (per day)
  - Area chart: new user registrations over time (per day, last 30 days)
  - Area chart: new jobs added per day (last 30 days, per platform)
  - Stat cards: total AI sessions, average ATS score, average match score
  - Area chart: AI sessions over time (last 30 days)
  - Subscription conversion card: free vs paid user ratio
  - Active users per day (last 7 days)
  - Scraper success rate per platform (last 10 cycles)

Metric Refresh

- Admin — clicks "Refresh" or metrics auto-refresh every 5 minutes
- System — invalidates cache, recomputes metrics, returns updated data

1.5 Validations

Security Validations

Check	Details
Authentication	Django admin session required — `is_staff=True` and `is_admin=True`
Role-based access	Monitoring views only accessible to admins
Read-only	Monitoring views do not mutate data

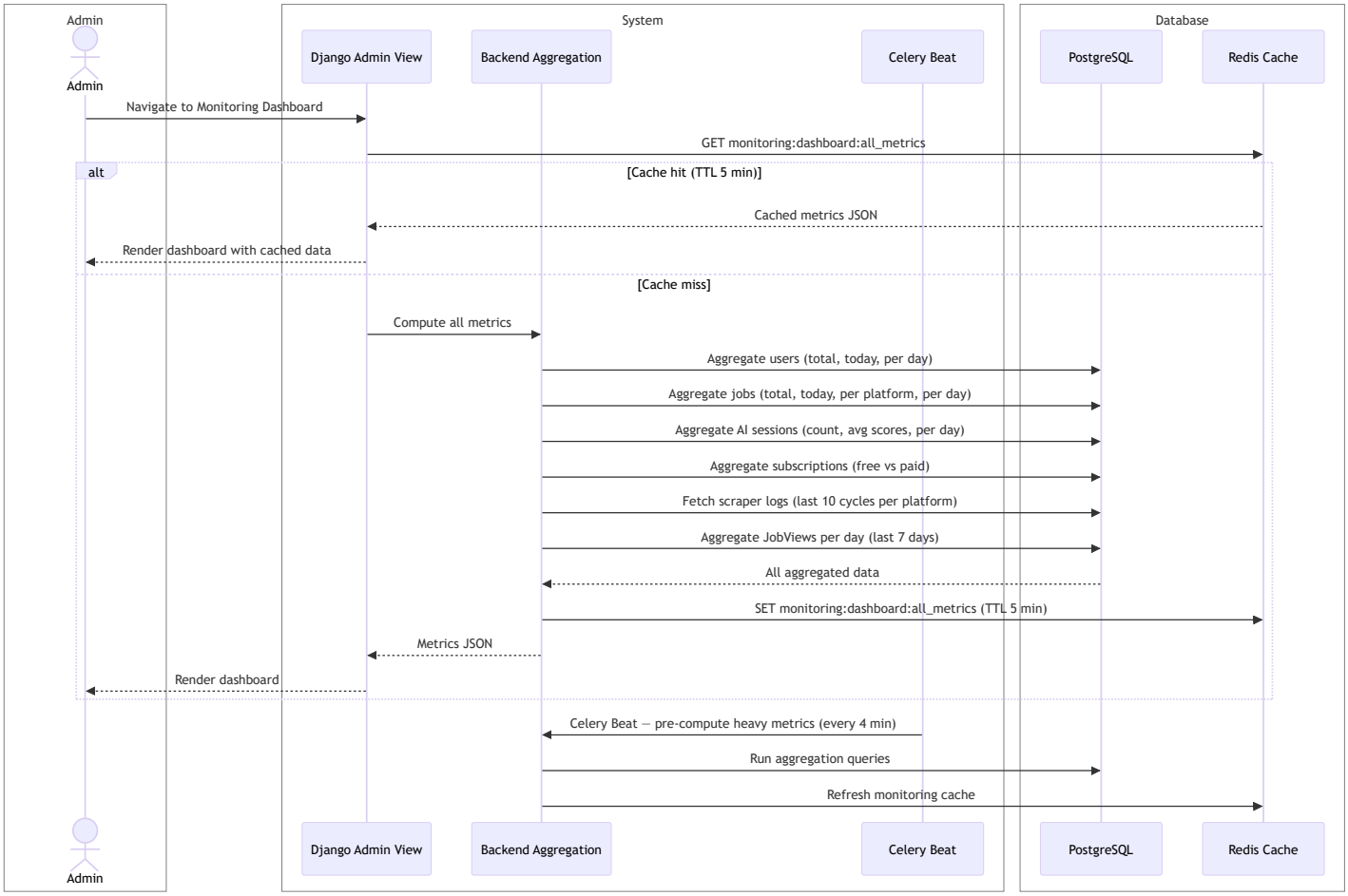
Error Handling

Scenario	System Response
DB aggregation query times out	Show last cached value with "Last updated: X minutes ago"
Scraper log missing	Show "No data" for that platform
Redis unavailable	Compute directly from DB, warn admin of degraded performance

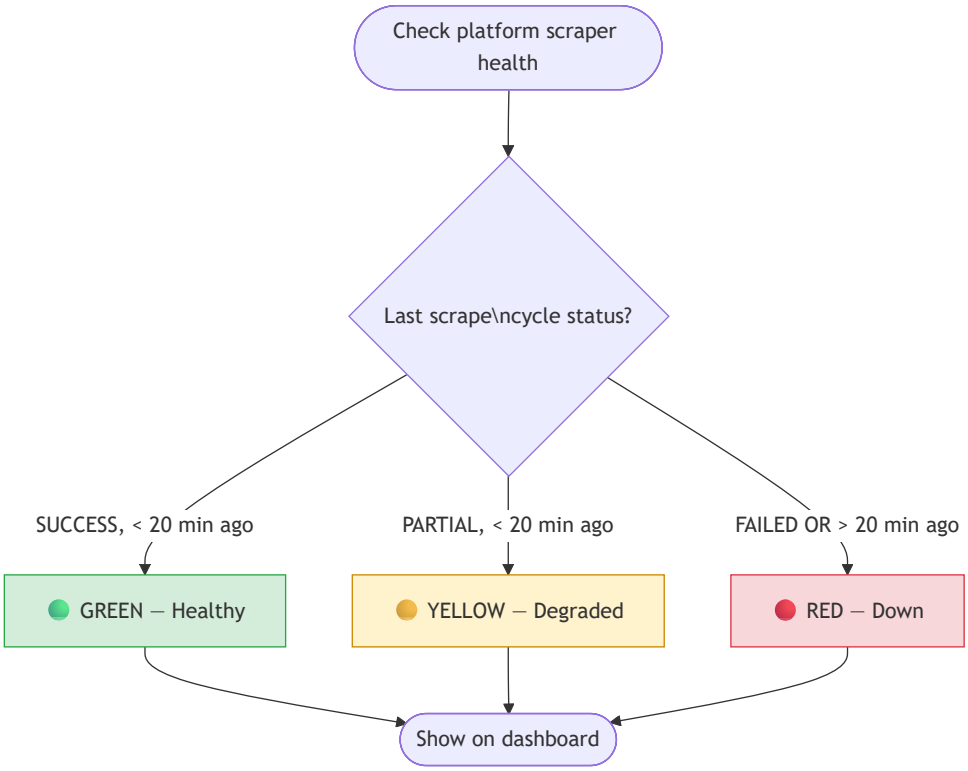
PART 2 — TECHNICAL

2.1 Diagrams

Sequence Diagram — Dashboard Load



Flowchart — Scraper Health Color Code Logic



2.2 Data Models

Model: `MetricSnapshot` *(optional — for historical trending)*

Purpose: Pre-aggregated daily metric snapshots to power fast historical charts without heavy live queries
Django app: `monitoring`

Field	Django Field Type	Required	Default	Notes
`id`	`UUIDField(primary_key=True)`	Auto	`uuid4`	PK
`date`	`DateField`	Yes	—	The day this snapshot covers. Indexed.
`metric_key`	`CharField(max_length=100)`	Yes	—	e.g. `new_users` , `new_jobs` , `jobs_viewed` , `ai_sessions`
`platform`	`CharField(max_length=20, null=True)`	No	`null`	Non-null for per-platform metrics
`value`	`FloatField`	Yes	—	The aggregated value for that day
`created_at`	`DateTimeField(auto_now_add=True)`	Auto	`now`	—

Unique constraint: `unique_together = [('date', 'metric_key', 'platform')]`

2.3 Metrics Reference

All metrics, their source models, and how they are computed:

Metric	Source	Computation
Total users	`User`	`User.objects.filter(status='ACTIVE').count()`
New users today	`User`	`User.objects.filter(created_at__date=today).count()`
New registrations per day (30d)	`User`	`GROUP BY DATE(created_at)` last 30 days

Metric	Source	Computation
Total jobs in system	`Job`	`Job.objects.filter(status='ACTIVE').count()`
New jobs today	`Job`	`Job.objects.filter(scraped_at__date=today).count()`
New jobs per day per platform (30d)	`Job`	`GROUP BY DATE(scraped_at), platform`
Jobs viewed (this week)	`JobView`	`GROUP BY DATE(viewed_at)` last 7 days
Total AI sessions	`AISession`	`AISession.objects.count()`
AI sessions per day (30d)	`AISession`	`GROUP BY DATE(created_at)`
Resumes checked by AI	`AISession`	`AISession.objects.filter(session_type='ATS_CHECK', status='COMPLETE').count()`
Avg ATS score (global)	`AISession`	`Avg('score') WHERE session_type='ATS_CHECK'`
Avg match score (global)	`AISession`	`Avg('score') WHERE session_type='JOB_MATCH'`
Active users per day (7d)	`JobView` or `AISession`	Distinct users with any activity per day
Scraper success rate	`ScrapeLog`	Last 10 logs per platform — `SUCCESS / total * 100`
Free vs paid users	`UserSubscription`	`GROUP BY plan.name`
Subscription conversion rate	`UserSubscription`	`paid_users / total_users * 100`

2.4 API Endpoints

Method	Endpoint	Auth	Role	Response	Description
`GET`	`/api/v1/admin/monitoring/`	Yes	Admin	`200` — full metrics JSON	All dashboard metrics
`GET`	`/api/v1/admin/monitoring/scrapers/`	Yes	Admin	`200` — scraper health per platform	Scraper status + last cycle details
`POST`	`/api/v1/admin/monitoring/refresh/`	Yes	Admin	`200`	Force cache refresh

2.5 Developer Notes

🔵 Backend Developer (Django)

- Create a `monitoring` Django app.
- Create a custom Django admin view (not a ModelAdmin — a plain `AdminSite` view or a custom `TemplateView` registered in the admin). Use `admin.site.register_view` or override `get_urls()` on a custom `AdminSite`.
- All aggregation queries should be wrapped in `try/except` — return last cached value if a query fails.
- Use Django ORM's `annotate`, `values`, `Count`, `Avg`, `TruncDate` for all aggregations. Example:

```
from django.db.models.functions import TruncDate
from django.db.models import Count

User.objects.annotate(day=TruncDate('created_at')) \
    .values('day') \
    .annotate(count=Count('id')) \
    .order_by('day')
```

- Celery Beat task `refresh_monitoring_cache`: runs every 4 minutes, pre-computes all metrics and sets them in Redis under `monitoring:dashboard:all_metrics` with TTL 5 minutes. This ensures the dashboard almost always hits cache.
- `MetricSnapshot` model: populate daily via a Celery Beat task at midnight. Used for charts that need historical data beyond 30 days. For the MVP, direct DB aggregation for the last 30 days is acceptable.
- Scraper health: fetch `ScrapeLog.objects.filter(platform=p).order_by('-started_at')[:10]` for each platform. Compute success rate. Determine health color: GREEN if latest is SUCCESS and `started_at > now - 20min`, YELLOW if PARTIAL, RED if FAILED or stale.
- Protect all monitoring endpoints with a custom permission: `IsAdminUser` (checks `user.is_admin=True`).

🟢 Frontend Developer (React)

- Monitoring dashboard is built inside Django admin — use Django's template system with a custom admin template.
- Embed a React-powered chart component via a `<script>` tag in the Django template, or use a pure JS charting library (Chart.js or Recharts via CDN) to avoid a full React build in admin.
- Charts needed:
  - **Area chart** — jobs viewed per day (last 7 days) — `Chart.js` area type
  - **Area chart** — new users per day (last 30 days)
  - **Area chart** — new jobs per day per platform (last 30 days) — multi-series, one per platform
  - **Area chart** — AI sessions per day (last 30 days)
  - **Bar chart** — scraper success rate per platform
  - **Donut chart** — free vs paid users
- Stat cards: simple HTML/CSS cards with large numbers — total users, total jobs, avg ATS score, conversion rate.
- Color-coded scraper health: green/yellow/red dot next to each platform name.
- Auto-refresh: `setInterval` every 5 minutes — re-fetch `/api/v1/admin/monitoring/` and update charts.
- "Refresh" button for manual refresh — calls `POST /api/v1/admin/monitoring/refresh/` then re-fetches.

🟡 Mobile Developer (Flutter)

N/A — monitoring dashboard is admin-only and web-based. No mobile implementation needed.

🟣 AI Engineer

- Avg ATS score and avg suitability score displayed on the dashboard are computed from `AISession` records — no AI involvement in the monitoring itself.
- Monitoring helps AI engineers track: how many sessions are running, average scores over time (useful for detecting model drift), and session failure rates.
- If the AI session failure rate exceeds a threshold (e.g. >10% in the last 24 hours), the monitoring dashboard should surface this as a RED alert in the AI health section.
- Recommendation: add a `monitoring:ai_health` Redis key that is updated by the AI Celery tasks themselves — increments a success/fail counter that the monitoring endpoint reads. This gives near-real-time AI health without a heavy DB query.

2.6 General Notes

- Consider adding email alerts to admin when scraper health goes RED for > 30 minutes